

Reductive groups of semisimple rank 1

SGA 3, Exposé XX

English draft

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About this draft

This is an unofficial English translation of Exposé XX, “Reductive groups of semisimple rank 1” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF Exp20-13oct24.pdf (35 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % `typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % `typo?:`. Scholarly proofreading remains outstanding.

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Exposé XX. Reductive groups of semisimple rank 1

by M. Demazure

1. Elementary systems. The groups U_α and $U_{-\alpha}$

(0)

Reminder 1.1. Let $S = \text{Spec}(k)$, where k is an algebraically closed field, and let G be a reductive S -group of semisimple rank 1, T a maximal torus (necessarily split) of G . One then has

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha},$$

where α and $-\alpha$ are the roots of G with respect to T . Moreover, there exist two monomorphisms of groups

$$p_\alpha : \mathbf{G}_{a,S} \longrightarrow G \quad \text{and} \quad p_{-\alpha} : \mathbf{G}_{a,S} \longrightarrow G$$

such that

$$tp_\alpha(x)t^{-1} = p(\alpha(t)x) \quad \text{and} \quad tp_{-\alpha}(x)t^{-1} = p_{-\alpha}(\alpha(t)^{-1}x),$$

for every $S' \longrightarrow S$ and all $t \in T(S')$, $x \in \mathbf{G}_a(S')$, and such that the morphism

$$\mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \longrightarrow G,$$

defined by $(y, t, x) \longmapsto p_{-\alpha}(y)tp_\alpha(x)$, is radicial and dominant (*Bible*, § 13.4, cor. 2 to th. 3).

Since the tangent map at the identity element is bijective, this morphism is also separable, hence birational; by Zariski's "Main Theorem" (EGA III₁, 4.4.9), it is therefore an open immersion.

Lemma 1.2. Let S be a scheme, G a group S -scheme, T a torus of G , Q a subtorus of T , α a character of T inducing on Q a character nontrivial on each fiber. Let $p_\alpha : \mathbf{G}_{a,S} \longrightarrow G$ (resp. $p_{-\alpha} : \mathbf{G}_{a,S} \longrightarrow G$) be a morphism of groups normalized by T with multiplier α (resp. $-\alpha$). Suppose that the morphism

$$u : \mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \longrightarrow G$$

defined set-theoretically by $u(y, t, x) = p_{-\alpha}(y)tp_\alpha(x)$ is an open immersion. Finally, let q be an integer ≥ 0 and

$$p : \mathbf{G}_{a,S} \longrightarrow G$$

a morphism of groups such that for every $S' \longrightarrow S$ and all $t \in Q(S')$, $x \in \mathbf{G}_a(S')$, one has

$$\text{int}(t)^q(p(x)) = p(\alpha(t)x).$$

There then exists a unique $\nu \in \mathbf{G}_a(S)$ such that $p(x) = p_\alpha(\nu x^q)$.

Indeed, let Ω be the image of u and $U = p^{-1}(\Omega)$. This is an open subset of $\mathbf{G}_{a,S}$ containing the zero section. For every section t of Q , the automorphism of $\mathbf{G}_{a,S}$ defined by multiplication by $\alpha(t)$ preserves U globally. One has $U = \mathbf{G}_{a,S}$; indeed, it suffices to check this when S is the spectrum of an algebraically closed field k ; then $\alpha : Q(k) \longrightarrow k^*$ is surjective, which proves at once that $U(k) \supset k^*$, hence $U = \mathbf{G}_{a,k}$. There therefore exist three morphisms

$$a : \mathbf{G}_{a,S} \longrightarrow \mathbf{G}_{a,S}, \quad b : \mathbf{G}_{a,S} \longrightarrow T, \quad c : \mathbf{G}_{a,S} \longrightarrow \mathbf{G}_{a,S},$$

(0)**N.D.E.** Version of 13/10/2024

such that

$$p(x) = p_{-\alpha}(a(x))b(x)p_{\alpha}(c(x)).$$

The condition on p becomes

$$\begin{aligned} a(\alpha(t)x) &= \alpha(t)^{-q}a(x), \\ b(\alpha(t)x) &= b(x), \\ c(\alpha(t)x) &= \alpha(t)^q c(x). \end{aligned}$$

For the same reason as above, for every $S' \rightarrow S$ and every $z \in \mathbf{G}_m(S')$, one therefore has

$$a(zx) = z^{-q}a(x), \quad b(zx) = b(x), \quad c(zx) = z^q c(x),$$

hence

$$z^q a(z) = a(1), \quad b(z) = b(1), \quad c(z) = z^q c(1).$$

Since $\mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}$, one has at once, for every $x \in \mathbf{G}_a(S')$, $S' \rightarrow S$:

$$\begin{aligned} x^q a(x) &= a(1) = a(0) = 0, & \text{whence } a &= 0, \\ c(x) &= x^q c(1) = \nu x^q, & \text{for some } \nu &\in \mathbf{G}_a(S), \\ b(x) &= b(1) = b(0) = e, & \text{whence } b &= e, \end{aligned}$$

which completes the proof.

Definition 1.3. Let S be a scheme. An *elementary S -system* is a triple (G, T, α) where

- (i) G is a reductive S -group of semisimple rank 1 (Exp. XIX 2.7),
- (ii) T is a maximal torus of G ,
- (iii) α is a root of G with respect to T (Exp. XIX 3.2).

One therefore has a direct sum decomposition (Exp. XIX 3.5)⁽¹⁾

$$\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^{\alpha} \oplus \mathfrak{g}^{-\alpha},$$

$\mathfrak{g}^{\alpha}$ and $\mathfrak{g}^{-\alpha}$ being locally free of rank one.

1.4. — If (G, T, α) is an elementary S -system, then $(G_{S'}, T_{S'}, \alpha_{S'})$ is an elementary S' -system for every $S' \rightarrow S$. If (G, T, α) is an elementary S -system, then $(G, T, -\alpha)$ is also one.

If S is a scheme, G a reductive S -group, T a maximal torus of G , α a root of G with respect to T , then (Exp. XIX 3.9) (Z_{α}, T, α) is an elementary S -system.

Let (G, T, α) be an elementary S -system. The invertible module $\mathfrak{g}^{\alpha}$ is canonically equipped with a T -module structure. One therefore also has a T -module structure on the vector bundle $W(\mathfrak{g}^{\alpha})$. On the other hand, the inner automorphisms of T define on G a structure of group with operator group T .

Theorem 1.5. Let (G, T, α) be an elementary S -system.

- (i) There exists a unique morphism of groups with operator group T

$$\exp : W(\mathfrak{g}^{\alpha}) \rightarrow G$$

which induces on Lie algebras the canonical morphism $\mathfrak{g}^{\alpha} \rightarrow \mathfrak{g}$.⁽²⁾

In other words, $\exp$ is the unique morphism satisfying the following conditions: for every $S' \rightarrow S$ and all $t \in T(S')$, $X, X' \in W(\mathfrak{g}^{\alpha})(S')$, one has

$$\begin{aligned} \exp(X + X') &= \exp(X) \exp(X'), \\ \text{int}(t)(\exp(X)) &= \exp(\alpha(t)X), \\ \underline{\text{Lie}}(\exp)(X) &= X. \end{aligned}$$

⁽¹⁾**N.D.E.** Of $\mathcal{O}_S$ -modules.

⁽²⁾**N.D.E.** We shall see later (Cor. 5.9) that $\exp$ is an isomorphism from $W(\mathfrak{g}^{\alpha})$ onto a closed subgroup of G .

(ii) If one defines similarly (in the elementary S -system $(G, T, -\alpha)$)

$$\exp : W(\mathfrak{g}^{-\alpha}) \longrightarrow G,$$

then the morphism

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^{\alpha}) \longrightarrow G$$

defined set-theoretically by $(Y, t, X) \mapsto \exp(Y) \cdot t \cdot \exp(X)$ is an open immersion.

Suppose we have proved the existence of the required morphisms $\exp$, and let us prove the other assertions of the theorem. Let us first prove (ii). Since both sides are of finite presentation and flat over S , it suffices to do so when S is the spectrum of an algebraically closed field (SGA 1, I 5.7 and VIII 5.5). Let then $S = \text{Spec } k$. Let $X \in \Gamma(S, \mathfrak{g}^{\alpha})^{\times}$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^{\times}$. It suffices to prove that the morphism

$$\mathbf{G}_{a,k} \times_k T \times_k \mathbf{G}_{a,k} \longrightarrow G, \quad (y, t, x) \mapsto \exp(yY)t \exp(xX)$$

is an open immersion. Now, by 1.1 and 1.2, there exist $a, b \in k$ such that

$$\exp(yY) = p_{-\alpha}(ay) \quad \text{and} \quad \exp(xX) = p_{\alpha}(bx).$$

Since $\exp : W(\mathfrak{g}^{-\alpha}) \longrightarrow G$ induces a monomorphism on Lie algebras, one has $a \neq 0$; similarly $b \neq 0$, and one is reduced to 1.1.

The uniqueness of the morphism $\exp$ can be proved locally on S ; one then reduces to the case where $\mathfrak{g}^{\alpha}$ and $\mathfrak{g}^{-\alpha}$ are free, and one need only apply 1.2 (with $Q = T$ and $q = 1$).

It therefore remains to prove the existence of the required morphism $\exp$. Note first that, by the theory of faithfully flat descent and the preceding uniqueness assertion, it suffices to prove this existence locally on S for the (fpqc) topology. By the usual arguments using finite presentation, one reduces to the case where S is noetherian, then to the case where it is local noetherian. By the preceding remark, one may therefore content oneself with proving the existence of the desired morphism $\exp$ when $S = \text{Spec}(A)$, A local noetherian complete with algebraically closed residue field k . Let then $p_0 : \mathbf{G}_{a,k} \longrightarrow G_k$ be a monomorphism of k -groups normalized by T_k with multiplier $\alpha_0 = \alpha \otimes_A k$ (one exists by 1.1). One knows (1.1 and 1.2) that the corresponding morphism $T_k \cdot_{\alpha_0} \mathbf{G}_{a,k} \longrightarrow G_k$ is an immersion, hence in particular a monomorphism. Let us provisionally admit the following two lemmas:

Lemma 1.6. *Let S be a scheme, G an S -group of finite presentation, T an S -torus, α a character nontrivial on each fiber of T , s_0 a point of S . Let*

$$f : T \cdot_{\alpha} \mathbf{G}_{a,S} \longrightarrow G$$

be a morphism of S -groups such that f_{s_0} is a monomorphism and the restriction of f to T is a monomorphism. There exists an open neighborhood U of s_0 such that $f|_U$ is a monomorphism.

Lemma 1.7. *Let A be a complete local noetherian ring with algebraically closed residue field k , (G, T, α) an elementary A -system, $p_0 : \mathbf{G}_{a,k} \longrightarrow G_k$ a morphism of k -groups normalized by T_k with multiplier $\alpha \otimes_A k$. There exists a morphism of groups $p : \mathbf{G}_{a,A} \longrightarrow G$ normalized by T with multiplier α .*

Let p be the morphism whose existence is asserted by 1.7. Let $f : T \cdot_{\alpha} \mathbf{G}_{a,S} \longrightarrow G$ be the corresponding morphism. It satisfies the hypotheses of 1.6, hence is a monomorphism; in particular p is a monomorphism. One then concludes by Exp. XIX 4.9.

Proof of 1.6. Let $\varepsilon : S \longrightarrow T \cdot_{\alpha} \mathbf{G}_{a,S}$ denote the unit section. Since f is unramified at $\varepsilon(s_0)$, it is so at $\varepsilon(s)$ for all s in an open neighborhood U of s_0 ; $f|_U$ is therefore unramified (Exp. X 3.5),⁽³⁾ hence its kernel $\text{Ker}(f)|_U$ is unramified over U . To prove that $f|_U$ is a monomorphism, it therefore suffices⁽⁴⁾ to prove that $\text{Ker}(f)|_U$ is radicial over U , which is a set-theoretic question. One is therefore reduced to proving:

⁽³⁾**N.D.E.** See also VI_B, 2.5.

⁽⁴⁾**N.D.E.** By EGA IV₄, 17.9.1.

Lemma 1.8. *Let k be an algebraically closed field; let N be an invariant subgroup of $T \cdot_\alpha \mathbf{G}_{a,k}$ (α a nontrivial character of the torus T), étale over k and such that $N \cap T = \{e\}$. Then $N = \{e\}$.*

One has $\text{int}(t')(t, x) = (t, \alpha(t')x)$. If (t, x) is a point of N , with $x \neq 0$, then (t, zx) is also a point of N for $z \in k^*$, and (t, x) is not isolated, hence N is not quasi-finite. One therefore has set-theoretically $N \subset T$, and the proof is finished.

Proof of 1.7. Let $\mathfrak{m}$ be the radical of A and $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, $n \geq 0$. Let us first show, by induction on n , that p_0 can be extended for each n to a morphism of S_n -groups

$$p_n : \mathbf{G}_{a,S_n} \longrightarrow G_{S_n}$$

normalized by T_{S_n} with multiplier α_{S_n} , the p_n satisfying in addition $p_{n+1} \times_{S_{n+1}} S_n = p_n$.

Let $H = T \cdot_\alpha \mathbf{G}_{a,S}$. The morphism $H_{S_n} \longrightarrow G_{S_n}$ defined by p_n is denoted f_n . Let us admit the following lemma:

Lemma 1.9. *If (G, T, α) is an elementary k -system, k algebraically closed, and $p : \mathbf{G}_{a,k} \longrightarrow G$ is a monomorphism normalized by T with multiplier α , one has*

$$H^2(T \cdot_\alpha \mathbf{G}_{a,k}, \mathfrak{g}) = 0.$$

(One makes $T \cdot_\alpha \mathbf{G}_{a,k}$ act on $\mathfrak{g}$ through the morphism $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow G$ defined by p , and the adjoint representation of G .)

Then, by Exp. III 2.8, f_n will extend to a morphism of S_{n+1} -groups

$$f'_{n+1} : H_{S_{n+1}} \longrightarrow G_{S_{n+1}}.$$

Now f'_{n+1} and the canonical immersion of $T_{S_{n+1}}$ into $G_{S_{n+1}}$ have the same restriction to T_{S_n} . By Exp. III 2.5, there exists an element $g \in G(S_{n+1})$ such that $g \times_{S_{n+1}} S_n = e$ and such that $f_{n+1} = \text{int}(g) \circ f'_{n+1}$ restricts to T_{n+1} by the canonical immersion of T_{n+1} . Let p_{n+1} be the restriction of f_{n+1} to $\mathbf{G}_{a,S_{n+1}}$. It is a morphism normalized by $T_{S_{n+1}}$ with multiplier $\alpha_{S_{n+1}}$, extending p_n .

One has therefore constructed a coherent system (f_n) , and one must now algebraize it. Now one has:

Lemma 1.10. *Let A be a complete local noetherian ring, $\mathfrak{m}$ its maximal ideal, $S = \text{Spec}(A)$, $S_n = \text{Spec}(A/\mathfrak{m}^{n+1})$, T an S -torus, α a nonzero character of T , X an affine S -scheme on which T acts. Let T act on $\mathbf{G}_{a,S}$ through α . Let q be an integer ≥ 0 , and let $(f_n)_{n \geq 0}$ be a coherent system of morphisms*

$$f_n : \mathbf{G}_{a,S_n}^q \longrightarrow X_{S_n}$$

of objects with operators T_{S_n} . There exists a unique morphism of objects with operators T

$$f : \mathbf{G}_{a,S}^q \longrightarrow X$$

inducing the f_n (compare with Exp. IX 7.1).

Corollary 1.11. *If X is a group with operator group T and the f_n are morphisms of groups, f is also one.*

It suffices to apply the uniqueness assertion of the lemma to the two morphisms $\mathbf{G}_{a,S}^{2q} \longrightarrow X$ deduced from f in the usual way.

Proof of 1.10. Suppose T split, which is moreover the case in the application of 1.10 to the proof of 1.5. One knows (Exp. I 4.7.3.1) that $X \longmapsto \mathcal{A}(X)$ gives an equivalence between the category of affine S -schemes equipped with an action of T and the opposite of the category of S -algebras graded of type $M = \text{Hom}_{S\text{-gr.}}(T, \mathbf{G}_{m,S})$.

One therefore has gradings

$$B = \mathcal{A}(X) = \coprod_{m \in M} B_m \quad \text{and} \quad C = \mathcal{A}(\mathbf{G}_{a,S}^q) = \coprod_{m \in M} C_m.$$

One sees at once that each C_m is *free of finite type* over A . (Indeed, $C_m = 0$ if m is not a multiple of α , and if $m = d\alpha$, C_m is isomorphic to the A -module of homogeneous polynomials of degree d in q variables.) Put

$$\begin{aligned}\widehat{B}_m &= \varprojlim_n B_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{C}_m &= \varprojlim_n C_m \otimes_A (A/\mathfrak{m}^{n+1}), \\ \widehat{B} &= \prod_{m \in M} \widehat{B}_m, \quad \widehat{C} = \prod_{m \in M} \widehat{C}_m.\end{aligned}$$

One then has canonical morphisms of algebras graded of type M

$$g_B : B \longrightarrow \widehat{B} \quad \text{and} \quad g_C : C \longrightarrow \widehat{C}.$$

It follows from the remark above that g_C is an isomorphism. Giving a coherent system (f_n) as in the statement is equivalent to giving a morphism of graded A -algebras

$$\widehat{F} : \widehat{B} \longrightarrow \widehat{C}.$$

Finding a morphism f as in the statement is equivalent to finding a morphism of graded A -algebras $F : B \longrightarrow C$ making the diagram commute:

$$\begin{array}{ccc} B & \xrightarrow{F} & C \\ g_B \downarrow & & \downarrow g_C \\ \widehat{B} & \xrightarrow{\widehat{F}} & \widehat{C}. \end{array}$$

Since g_C is an isomorphism, the existence and uniqueness of F are immediate. This proves 1.10.

To finish the proof of 1.5, it therefore remains only to prove 1.9.

1.12. — *Proof of 1.9.* One has $\mathfrak{g} = \mathfrak{t} \oplus \mathfrak{g}^\alpha \oplus \mathfrak{g}^{-\alpha}$. As explained in 1.9, consider $\mathfrak{g}$ as a $(T \cdot_\alpha \mathbf{G}_{a,k})$ -module. Clearly $\mathfrak{t} \oplus \mathfrak{g}^\alpha$ is a submodule of $\mathfrak{g}$, the quotient being isomorphic to $\mathfrak{g}^{-\alpha}$ as a k -vector space and even as a T -module. Clearly $\mathbf{G}_{a,k}$ acts trivially on this one-dimensional quotient (since every group morphism from $\mathbf{G}_{a,k}$ to $\mathbf{G}_{m,k}$ is trivial). Similarly, $\mathfrak{g}^\alpha$ is a submodule of $\mathfrak{t} \oplus \mathfrak{g}^\alpha$, the quotient being isomorphic to $\mathfrak{t}$ as a T -module, $\mathbf{G}_{a,k}$ acting trivially on it. To summarize:

Lemma 1.13. *Under the conditions of 1.9, $\mathfrak{g}$ has a composition series as a $(T \cdot_\alpha \mathbf{G}_{a,k})$ -module whose successive quotients are*

$$\mathfrak{g}^{-\alpha}, \quad \mathfrak{t}, \quad \mathfrak{g}^\alpha,$$

considered as $(T \cdot_\alpha \mathbf{G}_{a,k})$ -modules by the projection $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow T$.

One is therefore reduced to calculating the cohomology of $T \cdot_\alpha \mathbf{G}_{a,k}$ acting through the projection $T \cdot_\alpha \mathbf{G}_{a,k} \longrightarrow T$ and the character β of T (here $\beta = 0, \alpha$ or $-\alpha$) on $W(k)$.⁽⁵⁾ Denote by $k[x_1, \dots, x_n]$ the polynomial algebra over k in n variables, and by $k_q[x_1, \dots, x_n]$ the subspace of homogeneous polynomials of degree q .

Lemma 1.14. *With the preceding notations, one has $H^n(T \cdot_\alpha \mathbf{G}_{a,k}, k) = H^n(C_{\alpha, \beta}^*)$, where the complex $C_{\alpha, \beta}^*$ is defined by*

$$C_{\alpha, \beta}^n = \begin{cases} k_q[x_1, \dots, x_n] & \text{if } \beta = q\alpha, \text{ with } q \in \mathbf{N}^*; \\ 0 & \text{otherwise,} \end{cases}$$

⁽⁵⁾**N.D.E.** The following sentence has been added.

and

$$\begin{aligned} \delta f(x_1, x_2, \dots, x_{n+1}) &= f(x_2, \dots, x_{n+1}) \\ &+ \sum_{i=1}^n (-1)^i f(x_1, \dots, x_i x_{i+1}, \dots, x_{n+1}) \\ &+ (-1)^{n+1} f(x_1, \dots, x_n). \end{aligned}$$

Indeed, the functor $M \mapsto H^0(T, M)$ is exact on the category of T -modules (and the $H^q(T, -)$ vanish), by Exp. I 5.3.2. It follows, as in the usual case of group cohomology, that $H^n(T \cdot_\alpha \mathbf{G}_{a,k}, k)$ can be calculated as the n th cohomology group of the complex of cochains from $\mathbf{G}_{a,k}$ to k , invariant under T , i.e. satisfying

$$f(\alpha(t)x_1, \dots, \alpha(t)x_n) = \beta(t)f(x_1, \dots, x_n).$$

This indeed gives the announced complex.

To prove 1.9, it therefore suffices to prove that $H^2(C_{\alpha,\beta}^*) = 0$ for $\beta = 0, \alpha, -\alpha$, which is immediate.

Remark 1.15. One can explicitly calculate the groups $H^n(C_{\alpha,\beta}^*)$ for $\beta = q\alpha$ (see M. Lazard, *Lois de groupes et analyseurs*, Annales E.N.S., 1955). In particular, one finds $H^n(C_{\alpha,q\alpha}^*) = 0$ for $n > q$.

Notation 1.16. The image of the canonical immersion

$$W(\mathfrak{g}^{-\alpha}) \times_S T \times_S W(\mathfrak{g}^\alpha) \longrightarrow G$$

will be denoted Ω . It is an open subset of G containing the unit section. The image of

$$W(\mathfrak{g}^{-\alpha}), \quad \text{resp. } W(\mathfrak{g}^\alpha), \quad \text{resp. } W(\mathfrak{g}^{-\alpha}) \times_S T, \quad \text{resp. } T \times_S W(\mathfrak{g}^\alpha)$$

will be denoted⁽⁶⁾

$$U_{-\alpha}, \quad \text{resp. } U_\alpha, \quad \text{resp. } U_{-\alpha} \cdot T, \quad \text{resp. } T \cdot U_\alpha.$$

Then U_α (resp. $U_{-\alpha}$) is a subgroup of G canonically equipped with a vector bundle structure, and one has

$$\text{int}(t)(x) = x^{\alpha(t)} \quad (\text{resp. } x^{-\alpha(t)}),$$

for all $S' \longrightarrow S$, $t \in T(S')$, $x \in U_\alpha(S')$ (resp. $x \in U_{-\alpha}(S')$).

One has canonical isomorphisms

$$T \cdot U_\alpha \simeq T \cdot_\alpha U_\alpha \quad \text{and} \quad T \cdot U_{-\alpha} \simeq T \cdot_{-\alpha} U_{-\alpha}.$$

The open subset Ω is stable under $\text{int}(T)$: one has

$$\text{int}(t')(y \cdot t \cdot x) = y^{-\alpha(t')} \cdot t \cdot x^{\alpha(t')}.$$

Corollary 1.17. *One has $\mathcal{L}ie(U_\alpha/S) = \mathfrak{g}^\alpha$ and $\mathcal{L}ie(U_{-\alpha}/S) = \mathfrak{g}^{-\alpha}$. The isomorphisms*

$$W(\mathfrak{g}^\alpha) \xrightarrow[\sim]{\exp} U_\alpha \quad \text{and} \quad W(\mathfrak{g}^{-\alpha}) \xrightarrow[\sim]{\exp} U_{-\alpha}$$

are those of Exp. XIX 4.2.

Corollary 1.18. *The open subset Ω is relatively schematically dense in G (cf. XVIII, § 1).*

Clear by Exp. XVIII, 1.3.⁽⁷⁾

Corollary 1.19. *The center of G is $\text{Centr}(G) = \text{Ker}(\alpha)$. It is therefore a closed subgroup of G , of multiplicative type and of finite type.*

The second assertion follows from the first by Exp. IX 2.7. Let us therefore prove the latter. The inner automorphism defined by a section of $\text{Ker}(\alpha)$ acts trivially on Ω (last

⁽⁶⁾**N.D.E.** $P_{-\alpha}$ and P_α have been replaced by $U_{-\alpha}$ and U_α .

⁽⁷⁾**N.D.E.** See also EGA IV₃, 11.10.10.

formula of 1.16), hence on G by 1.18. Conversely, if $g \in G(S)$ centralizes G , it centralizes T and U_α , hence is a section of T (Exp. XIX 2.8) annihilating α ; since this also holds after every base change, one indeed has $\text{Centr}(G) = \text{Ker}(\alpha)$.

Corollary 1.20. *For there to exist a monomorphism $p_\alpha : \mathbf{G}_{a,S} \rightarrow G$ normalized by T with multiplier α , it is necessary and sufficient that the $\mathcal{O}_S$ -module $\mathfrak{g}^\alpha$ be free. More precisely, one has a bijection given by*

$$X_\alpha \mapsto (x \mapsto \exp(xX_\alpha)) \quad \text{and} \quad p_\alpha \mapsto \text{Lie}(p_\alpha)$$

between $\Gamma(S, \mathfrak{g}^\alpha)^\times$ and the set of monomorphisms p_α as above (which is also the set of isomorphisms of vector group schemes $\mathbf{G}_{a,S} \xrightarrow{\sim} U_\alpha$).⁽⁸⁾

Corollary 1.21. *The subgroups U_α and $U_{-\alpha}$ of G do not commute on any fiber.*

Indeed, if $(U_\alpha)_s$ and $(U_{-\alpha})_s$ commute, Ω_s is a subgroup of G_s , hence $\Omega_s = G_s$ ⁽⁹⁾ and G_s is solvable, which contradicts the hypothesis that G_s is reductive of semisimple rank 1.

2. Structure of elementary systems

Theorem 2.1. *Let S be a scheme, (G, T, α) an elementary S -system. There exist a morphism of $\mathcal{O}_S$ -modules*

$$\mathfrak{g}^\alpha \otimes_{\mathcal{O}_S} \mathfrak{g}^{-\alpha} \rightarrow \mathcal{O}_S, \quad (X, Y) \mapsto \langle X, Y \rangle,$$

and a morphism of S -groups

$$\alpha^* : \mathbf{G}_{m,S} \rightarrow T$$

such that for every $S' \rightarrow S$ and all $X \in \Gamma(S', \mathfrak{g}^\alpha \otimes \mathcal{O}_{S'})$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha} \otimes \mathcal{O}_{S'})$ one has the equivalence

$$\exp(X) \cdot \exp(Y) \in \Omega(S') \iff 1 + \langle X, Y \rangle \in \mathbf{G}_m(S'),$$

and under these conditions one has the formula:

$$(F) \quad \boxed{\exp(X) \cdot \exp(Y) = \exp\left(\frac{Y}{1 + \langle X, Y \rangle}\right) \alpha^*(1 + \langle X, Y \rangle) \exp\left(\frac{X}{1 + \langle X, Y \rangle}\right).}$$

Moreover, the morphisms $(X, Y) \mapsto \langle X, Y \rangle$ and α^ are uniquely determined, the first is an isomorphism, hence puts the modules $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ in duality, and one has $\alpha \circ \alpha^* = 2$ (squaring in $\mathbf{G}_{m,S}$).*

In view of the uniqueness assertions of the theorem, it suffices to give the proof locally on S . One may therefore suppose that $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ are free over S . Take then $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$, and put $p_\alpha(x) = \exp(xX)$, $p_{-\alpha}(y) = \exp(yY)$ for $x, y \in \mathbf{G}_a(S')$, $S' \rightarrow S$. By 1.5 and 1.21, it suffices to prove:

Proposition 2.2. *Let S be a scheme, G an S -group, T a torus of G , α a character of T nontrivial on each fiber, $p_\alpha : \mathbf{G}_{a,S} \rightarrow G$ (resp. $p_{-\alpha} : \mathbf{G}_{a,S} \rightarrow G$) a monomorphism of groups normalized by T with multiplier α (resp. $-\alpha$). Suppose that:*

- (i) *The morphism $\mathbf{G}_{a,S} \times_S T \times_S \mathbf{G}_{a,S} \rightarrow G$ defined by $(y, t, x) \mapsto p_{-\alpha}(y)tp_\alpha(x)$ is an open immersion. Denote its image by Ω .*
- (ii) *For every $s \in S$, $(p_\alpha)_s(\mathbf{G}_{a,\kappa(s)})$ and $(p_{-\alpha})_s(\mathbf{G}_{a,\kappa(s)})$ do not commute.*

Then there exist uniquely determined $a \in \mathbf{G}_a(S)$ and $\alpha^ \in \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$ with the following properties: for every $S' \rightarrow S$ and all $x, y \in \mathbf{G}_a(S')$, one has*

$$p_\alpha(x)p_{-\alpha}(y) \in \Omega(S') \iff 1 + axy \in \mathbf{G}_m(S'),$$

⁽⁸⁾**N.D.E.** Indeed, on the one hand, $\text{Lie}(\mathbf{G}_{a,S}) = \mathcal{O}_S$, and $\text{Lie}(p_\alpha)$ is an element of $\text{Hom}_{\mathcal{O}_S}(\mathcal{O}_S, \mathfrak{g}^\alpha) = \Gamma(S, \mathfrak{g}^\alpha)$.

⁽⁹⁾**N.D.E.** By 1.18.

and, under this condition, one has the formula

$$p_\alpha(x)p_{-\alpha}(y) = p_{-\alpha}\left(\frac{y}{1+axy}\right)\alpha^*(1+axy)p_\alpha\left(\frac{x}{1+axy}\right).$$

Moreover, a is invertible (i.e. $a \in \mathbf{G}_m(S)$) and $\alpha \circ \alpha^* = 2$.

Proof:

A) Consider the morphism

$$\mathbf{G}_{a,S}^2 \longrightarrow G$$

defined by $(x, y) \mapsto p_\alpha(x)p_{-\alpha}(y)$. Let U be the inverse image of Ω by this morphism. It is an open subset of $\mathbf{G}_{a,S}^2$ containing $0 \times_S \mathbf{G}_{a,S}$ and $\mathbf{G}_{a,S} \times_S 0$. There therefore exist uniquely determined morphisms of S -schemes

$$A : U \longrightarrow \mathbf{G}_{a,S}, \quad C : U \longrightarrow \mathbf{G}_{a,S}, \quad B : U \longrightarrow T$$

satisfying the set-theoretic relation:

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(A(u, v))B(u, v)p_\alpha(C(u, v)).$$

One immediately has the relations

$$\begin{aligned} A(0, v) &= v, & A(u, 0) &= 0, & C(u, 0) &= u, & C(0, v) &= 0, \\ B(u, 0) &= B(0, v) = e. \end{aligned}$$

Let S' be a separated S -scheme and $t \in T(S')$ a point of T . Since $\Omega_{S'}$ is stable under $\text{int}(t)$, by the last formula of 1.16, $U_{S'}$ is stable under the automorphism $(x, y) \mapsto (\alpha(t)x, \alpha(t)^{-1}y)$ of $\mathbf{G}_{a,S'}^2$, and one has the relations:

$$\begin{aligned} A(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)^{-1}A(u, v), \\ C(\alpha(t)u, \alpha(t)^{-1}v) &= \alpha(t)C(u, v), \\ B(\alpha(t)u, \alpha(t)^{-1}v) &= B(u, v). \end{aligned}$$

Since α is faithfully flat, one deduces that for every $S' \rightarrow S$ and every $z \in \mathbf{G}_m(S')$, $U_{S'}$ is stable under the transformation $(x, y) \mapsto (zx, z^{-1}y)$ and one has

$$\begin{aligned} A(zu, z^{-1}v) &= z^{-1}A(u, v), & C(zu, z^{-1}v) &= zC(u, v), \\ B(zu, z^{-1}v) &= B(u, v). \end{aligned}$$

Suppose first that v is invertible; putting $z = v$, one deduces that if (u, v) is a section of U , then $(uv, 1)$ is also one, and one has

$$A(uv, 1) = v^{-1}A(u, v), \quad B(uv, 1) = B(u, v).$$

Let then V be the open subset of $\mathbf{G}_{a,S}^2$ defined by⁽¹⁰⁾

$$(u, v) \in V(S') \iff (u, v), (uv, 1) \text{ and } (1, uv) \text{ belong to } U(S').$$

Since U is an open subset of $\mathbf{G}_{a,S}^2$ containing $0 \times_S \mathbf{G}_{a,S}$ and $\mathbf{G}_{a,S} \times_S 0$, V is a neighborhood of the zero section of $\mathbf{G}_{a,S}^2$, and we have just seen that the morphisms

$$\begin{aligned} (u, v) &\mapsto A(u, v) \quad \text{and} \quad (u, v) \mapsto vA(uv, 1), \\ \text{resp. } (u, v) &\mapsto B(u, v) \quad \text{and} \quad (u, v) \mapsto B(uv, 1) \end{aligned}$$

coincide in $V \cap (\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S})$. Since $\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}^2$, these morphisms therefore coincide in V .

One knows that $A(0, 1) = 1$; it follows that there exists an open subset W_1 of $\mathbf{G}_{a,S}$ containing the zero section, such that for every section x of W_1 , $A(x, 1)$ is invertible; putting $A(x, 1)^{-1} = F(x)$, one obtains that if $(u, v) \in V(S')$ ⁽¹¹⁾ and $uv \in W_1(S')$, $S' \rightarrow S$, then

⁽¹⁰⁾**N.D.E.** The condition “ $(1, uv) \in U(S')$ ” has been added.

⁽¹¹⁾**N.D.E.** Here and in what follows, $U(S')$ has been replaced by $V(S')$.

$A(u, v) = vA(uv, 1) = vF(uv)^{-1}$. Reasoning similarly for C , one obtains that there exist an open subset W_2 of $\mathbf{G}_{a,S}$ containing the zero section and an element $E^{(12)}$ of $\mathcal{O}(W_2)^\times$ such that $C(u, v) = uC(1, uv) = uE(uv)^{-1}$ if $(u, v) \in V(S')$ and $uv \in W_2(S')$. Consequently, putting $W = W_1 \cap W_2$, one obtains:

There exist an open subset W of $\mathbf{G}_{a,S}$ containing the zero section, and S -morphisms

$$\begin{aligned} F : W &\longrightarrow \mathbf{G}_{m,S}, & F(0) &= 1, \\ H : W &\longrightarrow T, & H(0) &= e, \\ E : W &\longrightarrow \mathbf{G}_{m,S}, & E(0) &= 1, \end{aligned}$$

such that if $(u, v) \in V(S')$ and $uv \in W(S')$, $S' \longrightarrow S$, one has

$$(+) \quad p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}(vF(uv)^{-1})H(uv)p_\alpha(uE(uv)^{-1}).$$

B) Let us now use associativity in G to write

$$p_\alpha(u)p_{-\alpha}(v)p_{-\alpha}(w) = p_\alpha(u)p_{-\alpha}(v+w).$$

There exists an open subset L of $\mathbf{G}_{a,S}^3$, containing the unit section, such that $(u, v, w) \in L(S')$ is equivalent to

$$\begin{aligned} (u, v) &\in V(S'), & (uE(uv)^{-1}, w) &\in V(S'), & (u, v+w) &\in V(S'), \\ uv &\in W(S'), & uwE(uv)^{-1} &\in W(S'), & u(v+w) &\in W(S'). \end{aligned}$$

Using formula (+), one writes at once, for $(u, v, w) \in L(S')$, the relations:

$$\begin{aligned} (1) \quad & E(uv + uw) = E(uwE(uv)^{-1})E(uv), \\ (2) \quad & H(uv + uw) = H(uwE(uv)^{-1})H(uv), \\ (3) \quad & (v+w)F(uv + uw)^{-1} = \alpha(H(uv)^{-1})wF(uwE(uv)^{-1})^{-1} \\ & \quad + vF(uv)^{-1}. \end{aligned}$$

It is immediate from the definition of L that $(1, 0, 0) \in L(S)$. Consider therefore

$$L \cap (1 \times_S \mathbf{G}_{a,S} \times_S \mathbf{G}_{a,S}) = 1 \times_S M;$$

M is an open subset of $\mathbf{G}_{a,S}^2$ containing the section $(0, 0)$, and, for $(v, w) \in M(S')$, one has $v, wE(v)^{-1}, v+w \in W(S')$ and

$$\begin{aligned} (1') \quad & E(v+w) = E(wE(v)^{-1})E(v), \\ (2') \quad & H(v+w) = H(wE(v)^{-1})H(v), \\ (3') \quad & (v+w)F(v+w)^{-1} = \alpha(H(v))^{-1}wF(wE(v)^{-1})^{-1} + vF(v)^{-1}. \end{aligned}$$

Finally, consider the morphism from M to $\mathbf{G}_{a,S}^2$ defined set-theoretically by $(v, w) \mapsto (v, wE(v)^{-1})$.⁽¹³⁾ It preserves the section $(0, 0)$ and induces an isomorphism from M onto an open subset N of $\mathbf{G}_{a,S}^2$ containing the zero section (the inverse isomorphism being given by $(x, y) \mapsto (x, yE(x))$).⁽¹⁴⁾ One has therefore proved the following assertion:

There exists an open subset N of $\mathbf{G}_{a,S}^2$ containing the zero section such that if $(x, y) \in N(S')$, then x, y and $x + yE(x)$ ⁽¹⁵⁾ belong to $W(S')$ and:

$$\begin{aligned} (1'') \quad & E(x + yE(x)) = E(x)E(y), \\ (2'') \quad & H(x + yE(x)) = H(x)H(y), \\ (3'') \quad & (x + yE(x))F(x + yE(x))^{-1} = xF(x)^{-1} + \alpha(H(x))^{-1}yE(x)F(y)^{-1}. \end{aligned}$$

⁽¹²⁾**N.D.E.** The element denoted G in the original has been denoted E , since G already denotes the S -group under consideration.

⁽¹³⁾**N.D.E.** The following has been corrected.

⁽¹⁴⁾**N.D.E.** I.e. one has made the “change of variables” $x = v, y = wE(v)^{-1}$, that is, $v = x, w = yE(x)$.

⁽¹⁵⁾**N.D.E.** $yE(x)$ has been corrected to $x + yE(x)$.

C) Reasoning similarly with associativity on the left, one proves the following assertion:⁽¹⁶⁾

There exists an open subset N' of $\mathbf{G}_{a,S}^2$ containing the zero section such that if $(x, y) \in N'(S')$, then x, y and $x + yF(x)$ ⁽¹⁷⁾ belong to $W(S')$, and

$$(4'') \quad F(x + yF(x)) = F(x)F(y),$$

$$(5'') \quad H(x + yF(x)) = H(x)H(y),$$

$$(6'') \quad (x + yF(x))E(x + yF(x))^{-1} = xE(x)^{-1} + \alpha(H(x))^{-1}yF(x)E(y)^{-1}.$$

We are therefore led to solve the “functional equation” (1'').

Lemma 2.3. *Let S be a scheme, W an open subset of $\mathbf{G}_{a,S}$ containing the unit section, $F : W \rightarrow \mathbf{G}_{m,S}$ an S -morphism. Suppose that $F(0) = 1$ and that there exists an open subset N of $\mathbf{G}_{a,S}^2$ containing the zero section such that for $(x, y) \in N(S')$, x, y and $x + yF(x)$ ⁽¹⁷⁾ belong to $W(S')$ and one has:*

$$(\dagger) \quad F(x + yF(x)) = F(x)F(y).$$

(i) *If S is the spectrum of a field k , there exists $a \in k$ such that $F(x) = 1 + ax$.*

(ii) *If $a = F'(0) \in \Gamma(S, \mathcal{O}_S)$ is invertible, then $F(x) = 1 + ax$.*

By the hypotheses, we can differentiate the given equation at $x = 0$ (resp. at $y = 0$), and find that

$$(*) \quad F'(y)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (0, y) \in N(S'),$$

resp.

$$F'(x)F(x) = F(x)F'(0) \quad \text{for } (x, 0) \in N(S').$$

Since F takes values in $\mathbf{G}_m$, the second relation gives

$$(*') \quad F'(x) = F'(0) \quad \text{for } (x, 0) \in N(S');$$

whence, by the first,

$$F'(0)(1 + yF'(0)) = F'(0)F(y) \quad \text{for } (y, 0), (0, y) \in N(S').$$

If $a = F'(0)$ is invertible, this gives

$$F(y) = 1 + ay,$$

for y a section of an open subset of W containing the unit section, hence schematically dense in W , which proves (ii). This also proves (i) when $F'(0) \neq 0$.

If $F'(0) = 0$, then, by (*'), $F'(x) = 0$ when x is “near 0”, hence $F' = 0$ by schematic density. If k has characteristic 0, F is a rational fraction with zero derivative, hence constant and equal to $F(0) = 1$.

If k has characteristic p and F is not constant,⁽¹⁸⁾ there exist an integer $n > 0$ and a rational fraction $F_1 \in k(X)$ such that $F'_1(X) \neq 0$ and

$$F(X) = F_1(X^{p^n}) = F_1(X)^{p^n}.$$

Substituting in the functional equation, one finds

$$(\dagger_1) \quad F_1(x + yF_1(x)^{p^n}) = F_1(x)F_1(y).$$

Differentiating at $x = 0$, one finds

$$(*_1) \quad F'_1(y) = F'_1(0)F_1(y),$$

and differentiating $(\dagger_1)$ at $y = 0$, one obtains

$$(*'_1) \quad F'_1(x)F_1(x)^{p^n} = F_1(x)F'_1(0).$$

⁽¹⁶⁾**N.D.E.** I.e. one writes the equalities resulting from $p_\alpha(t)p_\alpha(u)p_{-\alpha}(v) = p_\alpha(t+u)p_{-\alpha}(v)$, and puts $v = 1$ and $x = u$, $t = yF(u)$ (i.e. $y = tF(u)^{-1}$).

⁽¹⁷⁾**N.D.E.** $yF(x)$ has been corrected to $x + yF(x)$.

⁽¹⁸⁾**N.D.E.** The following has been corrected.

Since, by hypothesis, $F'_1(X)$ is an invertible element of $k(X)$, one deduces from these two equalities that

$$F_1(X)^{p^n} = 1,$$

hence F_1 is a constant, contradicting the initial hypothesis. This shows that F is constant and equal to $1 = F(0)$.

D) Suppose that S is the spectrum of a field. If $F'(0) = 0$, then $F = 1$. Formula (5'') then gives $H(x+y) = H(x)H(y)$, which shows that H extends to a morphism of groups $\mathbf{G}_{a,S} \rightarrow T$ (Exp. XVIII 2.3), which is necessarily constant with value e . On the other hand, by lemma 2.3, one will also have $E(x) = 1 + bx$ for some $b \in k$. But then (6'') gives, for $(x, y) \in N(S')$,

$$(x+y)E(x+y)^{-1} = xE(x)^{-1} + yE(y)^{-1},$$

hence,⁽¹⁹⁾ by Exp. XVIII 2.3 again, $x \mapsto xE(x)^{-1}$ extends to a morphism of k -groups $\mathbf{G}_{a,k} \rightarrow \mathbf{G}_{a,k}$, hence $x/(1+bx) = cx$ for some $c \in k$, whence $b = 0$ (and $c = 1$).

This shows that F, H, E are constant with values $(1, e, 1)$ in a neighborhood of the unit section, hence everywhere, which by (+) shows that U_α and $U_{-\alpha}$ commute, contrary to hypothesis (ii).

If S is now arbitrary, one has therefore proved that $F'(0)$ vanishes on no fiber, hence is invertible. The same is clearly true of $E'(0)$, which, by lemma 2.3, shows that there exist $a, b \in \mathbf{G}_m(S)$ such that

$$(\diamond_1) \quad F(x) = 1 + ax, \quad E(x) = 1 + bx, \quad \text{for } x \in W(S').$$

E) The rest is now easy. Substituting the preceding results in (3''), one finds

$$y\alpha(H(x))(1+ay) = y(1+ax+ay(1+bx))(1+ax).$$

This formula is valid for every section (x, y) of N . But since $\mathbf{G}_{a,S} \times_S \mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}^2$, one deduces

$$(1+ay)\alpha(H(x)) = (1+ax+ay(1+bx))(1+ax).$$

Putting $y = 0$ gives $\alpha(H(x)) = (1+ax)^2$. Substituting this in the preceding equality, one finds⁽²⁰⁾

$$a^2xy = abxy.$$

Since $\mathbf{G}_{m,S}$ is schematically dense in $\mathbf{G}_{a,S}$, one deduces $a^2 = ab$, whence, since a is invertible,

$$(\diamond_2) \quad a = b \quad \text{and} \quad \alpha(H(x)) = (1+ax)^2.$$

Since a is invertible, $x \mapsto 1+ax$ is an automorphism of $\mathbf{G}_{a,S}$; one can therefore find an open subset W' of $\mathbf{G}_{a,S}$ containing the section 1 and a morphism

$$P : W' \rightarrow T$$

such that $P(1+ax) = H(x)$.⁽²¹⁾

Substituting in relation (2'), one finds at once, for $(x, y) \in N(S')$,

$$P(1+ax+ay) = P\left(\frac{1+ax+ay}{1+ax}\right)P(1+ax),$$

which proves that there exists an open neighborhood of 1 in $\mathbf{G}_{m,S}$ such that for x and y in this neighborhood one has $P(x)P(y) = P(xy)$. By Exp. XVIII 2.3, there exists a morphism of groups

$$(\diamond_3) \quad \alpha^* : \mathbf{G}_{m,S} \rightarrow T$$

⁽¹⁹⁾**N.D.E.** The following sentence has been added. One can also see by a direct calculation that the preceding equality implies $0 = xyb(2 + (x+y)b)$, whence $0 = b(2 + (x+y)b)$, and finally $b = 0$.

⁽²⁰⁾**N.D.E.** In the preceding three equalities, the original has been corrected by replacing the factor $(1+bx)$ on the right by $1+ax$. Substituting the third equality in the second and taking account of the invertibility of $1+ax$, one indeed obtains $a^2xy = abxy$.

⁽²¹⁾**N.D.E.** I.e. one has made the change of variables $x' = 1+ax$, that is, $x = (x' - 1)/a$.

extending P . Since $\alpha(H(x)) = (1 + ax)^2$ in a neighborhood of the zero section, one has $\alpha(\alpha^*(z)) = z^2$ in a neighborhood of the section 1, hence

$$(\diamond_4) \quad \alpha \circ \alpha^* = 2.$$

F) ⁽²²⁾Collecting results (+) and $(\diamond_1 - \diamond_4)$, one sees that there exist $a \in \mathbf{G}_m(S)$ and $\alpha^* \in \text{Hom}_{S\text{-gr.}}(\mathbf{G}_{m,S}, T)$ such that $\alpha \circ \alpha^* = 2$, and such that, if $(u, v) \in V(S')$ and $uv \in W(S')$, then $1 + auv$ is invertible and

$$p_\alpha(u)p_{-\alpha}(v) = p_{-\alpha}\left(\frac{v}{1 + auv}\right)\alpha^*(1 + auv)p_\alpha\left(\frac{u}{1 + auv}\right).$$

Consider the open subset V' of $\mathbf{G}_{a,S}^2$ defined by “ $1 + auv$ invertible”, i.e. $V' = (\mathbf{G}_{a,S}^2)_f$, where $f(u, v) = 1 + auv$. Both sides of the preceding formula define morphisms from V' to G which coincide in a neighborhood of the zero section, hence coincide in V' . The preceding formula is therefore valid for every section (u, v) of V' . In particular, it follows that $V' \subset U$, where U is the open subset introduced at the beginning of A).

Let us prove that $U = V'$. Returning to the notations of A), one has a morphism

$$A : U \longrightarrow \mathbf{G}_{a,S}$$

which, on V' , is defined by $A(u, v) = v(1 + auv)^{-1}$. To show that $U = V'$, which is a set-theoretic question, one reduces to the case where S is the spectrum of a field k , hence to the following obvious assertion: the domain of definition of the rational map $\mathbf{G}_{a,k}^2 \longrightarrow \mathbf{G}_{a,k}$ defined by the rational fraction $Y/(1 + aXY)$ is the open subset defined by the function $1 + aXY$.

G) One has therefore proved the existence of a and α^* , as well as the two supplementary properties announced. It remains to prove uniqueness. Let therefore a' and $\alpha^{*'} also satisfy the required conditions. If $u, v \in \mathbf{G}_a(S')^2$, one has at once:$

$$1 + auv \text{ invertible} \Rightarrow 1 + a'uv \text{ invertible and } \frac{v}{1 + auv} = \frac{v}{1 + a'uv};$$

one therefore has for every section u of $\mathbf{G}_a(S')$

$$1 + au \text{ invertible} \Rightarrow 1 + au = 1 + a'u,$$

which proves at once that $a = a'$.

With the same notations, one then has

$$1 + au \text{ invertible} \Rightarrow \alpha^*(1 + au) = \alpha^{*'}(1 + au),$$

hence also $\alpha^* = \alpha^{*'}$.

Corollary 2.4. *Let $\exp(Y)t\exp(X)$ and $\exp(Y')t'\exp(X')$ be two elements of $\Omega(S')$. Then their product is in $\Omega(S')$ if and only if $u = 1 + \langle X, Y' \rangle$ is invertible, and one then has*

$$(F') \quad \begin{aligned} \exp(Y)t\exp(X) \cdot \exp(Y')t'\exp(X') &= \exp(Y + u^{-1}\alpha(t)^{-1}Y') \\ &\quad \cdot tt'\alpha^*(u) \cdot \exp(u^{-1}\alpha(t')^{-1}X + X'). \end{aligned}$$

Remark 2.5. One can also write formula (F) of theorem 2.1 without using the morphisms $\exp$. Indeed, transporting by these morphisms the duality $\mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} \longrightarrow \mathcal{O}_S$, one obtains a canonical pairing of vector bundles:

$$U_\alpha \times_S U_{-\alpha} \longrightarrow \mathbf{G}_{a,S},$$

which we shall again denote by $(x, y) \longmapsto \langle x, y \rangle$. One therefore has

$$\langle \exp X, \exp Y \rangle = \langle X, Y \rangle.$$

⁽²²⁾**N.D.E.** The following has been slightly modified, since an open subset V has already been introduced in A).

If $x \in U_\alpha(S')$, $y \in U_{-\alpha}(S')$ and $1 + \langle x, y \rangle \in \mathbf{G}_m(S')$, one has

$$(F) \quad x \cdot y = y^{(1+\langle x, y \rangle)^{-1}} \cdot \alpha^*(1 + \langle x, y \rangle) \cdot x^{(1+\langle x, y \rangle)^{-1}}.$$

Corollary 2.6. *The pairing*

$$W(\mathfrak{g}^\alpha) \times_S W(\mathfrak{g}^{-\alpha}) \longrightarrow \mathbf{G}_{a,S}$$

defines a pairing of principal bundles under $\mathbf{G}_{m,S}$

$$W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times \longrightarrow \mathbf{G}_{m,S}.$$

This pairing will be denoted $(X, Y) \mapsto \langle X, Y \rangle$, or more simply $(X, Y) \mapsto XY$.

For every section $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$, there therefore exists a unique section X^{-1} of $\Gamma(S, \mathfrak{g}^{-\alpha})^\times$ such that $XX^{-1} = 1$. One has $(zX)^{-1} = z^{-1}X^{-1}$. The morphism

$$s : W(\mathfrak{g}^\alpha)^\times \longrightarrow W(\mathfrak{g}^{-\alpha})^\times$$

thus defined is therefore an isomorphism of schemes, compatible with the isomorphism $s : z \mapsto z^{-1}$ on the operator groups.

Definition 2.6.1. One will say that X and $s(X) = X^{-1}$ are *paired*.

Apply corollary 2.4 to $Y = 0 = X'$ and $Y' = aX^{-1}$, $a \in \mathcal{O}_S(S)$. Then $u = 1 + a$ and $u^{-1}Y' = u^{-1}(u - 1)X^{-1} = (1 - u^{-1})X^{-1}$, whence:

Corollary 2.7. *Let $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and $u \in \Gamma(S, \mathcal{O}_S)^\times$. One has*

$$\alpha^*(u) = \exp((u^{-1} - 1)X^{-1}) \exp(X) \exp((u - 1)X^{-1}) \exp(-u^{-1}X).$$

Definition 2.8. The morphism α^* is called the *coroot* associated with the root α .

Remark 2.9. If (G, T, α) is an elementary S -system, $(G, T, -\alpha)$ is also one. One therefore has by theorem 2.1 a duality between $\mathfrak{g}^{-\alpha}$ and $\mathfrak{g}^\alpha$, and a coroot $(-\alpha)^*$. Taking the inverse of formula (F), one proves at once

$$\langle X, Y \rangle = \langle Y, X \rangle, \quad (-\alpha)^* = -\alpha^*.$$

Let us now pass to the Lie algebra of G . The root α and the coroot α^* define the linear forms

$$\mathcal{O}_S \xrightarrow{\bar{\alpha}^*} \mathfrak{t} \xrightarrow{\bar{\alpha}} \mathcal{O}_S.$$

One puts $H_\alpha = \bar{\alpha}^*(1)$. One calls $\bar{\alpha}$ the *infinitesimal root* associated with α , and H_α the corresponding *infinitesimal coroot*.

Lemma 2.10. *Let $S' \longrightarrow S$, and $X, X' \in W(\mathfrak{g}^\alpha)(S')$, $H \in W(\mathfrak{t})(S')$, $Y, Y' \in W(\mathfrak{g}^{-\alpha})(S')$, $t \in T(S')$. One has*

- (1) $\text{Ad}(t)H = H, \quad \text{Ad}(t)X = \alpha(t)X, \quad \text{Ad}(t)Y = \alpha(t)^{-1}Y.$
- (2) $\begin{cases} \text{Ad}(\exp(X))H = H - \bar{\alpha}(H)X, & \text{Ad}(\exp(X))X' = X', \\ \text{Ad}(\exp(X))Y = Y + \langle X, Y \rangle H_\alpha - \langle X, Y \rangle X. \end{cases}$
- (2') $\begin{cases} \text{Ad}(\exp(Y))H = H + \bar{\alpha}(H)Y, & \text{Ad}(\exp(Y))Y' = Y', \\ \text{Ad}(\exp(Y))X = X + \langle X, Y \rangle H_{-\alpha} - \langle X, Y \rangle Y. \end{cases}$
- (3) $[H, X] = \bar{\alpha}(H)X, \quad [H, Y] = -\bar{\alpha}(H)Y, \quad [X, Y] = \langle X, Y \rangle H_\alpha.$
- (4) $H_{-\alpha} = -H_\alpha.$
- (5) $\bar{\alpha}(H_\alpha) = 2.$

The proof of these various formulas is either trivial or an immediate consequence of formula (F) of 2.1.

Corollary 2.11. *Suppose that H_α is nonzero on every fiber (which is in particular the case if 2 is invertible on S , by (5)). Then $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ are paired if and only if $[X_\alpha, X_{-\alpha}] = H_\alpha$.*

2.12. — Let (G, T, α) be an elementary S -system. We know (1.19) that the center of G is $\text{Centr}(G) = \text{Ker}(\alpha)$, a group of multiplicative type and of finite type. If Q is a subgroup of multiplicative type of $\text{Centr}(G)$, the quotient G/Q is affine over S (Exp. IX 2.5), smooth over S (Exp. VI_B 9.2), with connected reductive fibers of semisimple rank 1 (Exp. XIX 1.8).

Put $G' = G/Q$; it is a reductive S -group of semisimple rank 1; $T' = T/Q$ is a maximal torus of it. The open subset $U_{-\alpha} \cdot T \cdot U_\alpha$ of G is stable under Q , and one sees at once that the quotient is isomorphic to $U_{-\alpha} \times_S (T/Q) \times_S U_\alpha$. If α' denotes the character of T' induced by α , it follows that the derived morphism of the canonical morphism $G \rightarrow G'$ induces isomorphisms

$$\mathfrak{g}^\alpha \xrightarrow{\sim} \mathfrak{g}'^{\alpha'} \quad \text{and} \quad \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}.$$

In particular, α' is a root of G' with respect to T' . Thus, denoting by α/Q the character $T/Q \rightarrow \mathbf{G}_{m,S}$ induced by α , one has:

Lemma 2.13. *If Q is a subgroup of multiplicative type of $\text{Ker}(\alpha)$, then*

$$(G/Q, T/Q, \alpha/Q)$$

is an elementary system.

Lemma 2.14. *Under the preceding conditions, the following diagrams commute:*

$$\begin{array}{ccccc} W(\mathfrak{g}^\alpha) & \xrightarrow{\text{exp}} & G & \xleftarrow{\text{exp}} & W(\mathfrak{g}^{-\alpha}) \\ \text{can} \downarrow \simeq & & \downarrow \text{can} & & \simeq \downarrow \text{can} \\ W(\mathfrak{g}'^{\alpha'}) & \xrightarrow{\text{exp}} & G' & \xleftarrow{\text{exp}} & W(\mathfrak{g}'^{-\alpha'}) \\ \\ \mathfrak{g}^\alpha \otimes \mathfrak{g}^{-\alpha} & \xrightarrow{\sim} & \mathcal{O}_S & & \\ \text{can} \downarrow \simeq & & \downarrow \text{id} & & \\ \mathfrak{g}'^{\alpha'} \otimes \mathfrak{g}'^{-\alpha'} & \xrightarrow{\sim} & \mathcal{O}_S & & \\ \\ \begin{array}{ccc} & T & \\ \alpha^* \nearrow & \downarrow \text{can} & \searrow \alpha \\ \mathbf{G}_{m,S} & & \mathbf{G}_{m,S} \\ \alpha'^* \searrow & \downarrow & \nearrow \alpha' \\ & T' & \end{array} & & \end{array}.$$

3. The Weyl group

Notation 3.0. ⁽²³⁾If (G, T, α) is an elementary S -system, one puts

$$N = \underline{\text{Norm}}_G(T), \quad W = \underline{\text{Norm}}_G(T)/T$$

(cf. Exp. XIX 6.3); N is a closed subgroup of G , smooth over S . One denotes by $N^\times = N - T$ the open subscheme of N induced on the complement of T .⁽²⁴⁾ Let R be the (unique) maximal torus of $\text{Ker}(\alpha)$, and T' the image of $\alpha^* : \mathbf{G}_{m,S} \rightarrow T$, which is a one-dimensional subtorus of T .

⁽²³⁾**N.D.E.** The numbering 3.0 has been added for subsequent references.

⁽²⁴⁾**N.D.E.** Q has been replaced by the more suggestive notation $N^\times$.

The morphism

$$T' \times_S R \longrightarrow T$$

induced by the product in T is surjective (hence faithfully flat); indeed, one reduces to checking this on geometric fibers, and it follows at once from the formula $\alpha \circ \alpha^* = 2$.

Theorem 3.1. *With the preceding notations:*

- (i) W is isomorphic to the constant group $(\mathbf{Z}/2\mathbf{Z})_S$.
- (ii) $N^\times$ is a locally trivial principal homogeneous bundle under T , on the left by the law $(t, q) \mapsto tq$ (resp. on the right by the law $(q, t) \mapsto qt$).
- (iii) One has the formula

$$\text{int}(w)t = t \cdot \alpha^*(\alpha(t)^{-1})$$

for $w \in N^\times(S')$, $t \in T(S')$, $S' \longrightarrow S$. In the decomposition $T_{S'} = T'_{S'} \cdot R_{S'}$, $\text{int}(w)$ induces the identity on $R_{S'}$ and reflection on $T'_{S'}$. One has the relations

$$\alpha \circ \text{int}(w) = \alpha^{-1}, \quad \text{int}(w) \circ \alpha^* = (\alpha^*)^{-1}.$$

- (iv) For $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \longrightarrow S$, put

$$w_\alpha(X) = \exp(X) \exp(-X^{-1}) \exp(X).$$

Then $w_\alpha(X) \in N^\times(S')$, and the morphism $w_\alpha : W(\mathfrak{g}^\alpha)^\times \longrightarrow N^\times$ thus defined satisfies

$$w_\alpha(zX) = \alpha^*(z)w_\alpha(X) = w_\alpha(X)\alpha^*(z)^{-1},$$

for $z \in \mathbf{G}_m(S')$, $X \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \longrightarrow S$.

- (v) ⁽²⁵⁾For $X, Y \in W(\mathfrak{g}^\alpha)^\times(S')$, $S' \longrightarrow S$, with the notations of 2.6, one has

$$w_\alpha(X)w_\alpha(Y) = \alpha^\vee(-XY^{-1}).$$

In particular,

$$w_\alpha(X)^2 = \alpha^*(-1) \in {}_2T(S) \cap \underline{\text{Centr}}(G)(S),$$

$$w_\alpha(X)^{-1} = w_\alpha(-X) = \alpha^*(-1)w_\alpha(X).$$

- (vi) If one similarly defines, for $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$,

$$w_{-\alpha}(Y) = \exp(Y) \exp(-Y^{-1}) \exp(Y),$$

one has (in addition to formulas analogous to the preceding ones)

$$w_{-\alpha}(X^{-1}) = w_\alpha(X)^{-1} = w_\alpha(-X),$$

$$w_\alpha(X)w_{-\alpha}(Y) = \alpha^*(XY).$$

Proof. (i) was already seen in Exp. XIX 2.4; it follows at once that $N^\times$ is indeed a principal homogeneous bundle under T for the laws defined in (ii); its local triviality (for the Zariski topology) follows, in particular, from (iv).

Let us prove (iii); if $w \in N^\times(S)$, clearly $\alpha \circ \text{int}(w)$ is a root of G with respect to T , hence is locally equal to α or $-\alpha$; since on each fiber it is $-\alpha$ (*Bible*, 12-05, proof of the cor. to prop. 1), one has $\alpha \circ \text{int}(w) = -\alpha$. By transport of structure, one deduces

$$-\alpha^* = \text{int}(w)^{-1} \circ \alpha^* = \text{int}(w) \circ \alpha^*,$$

since $\text{int}(w)^2 = \text{int}(w^2)$ and w^2 is a section of T . Thus $\text{int}(w)$ induces reflection on T' ; since R is central, $\text{int}(w)$ induces the identity on R . The formula of (iii) defines a morphism $T \longrightarrow T$ with the same properties, hence coincides with $\text{int}(w)$.

Let us prove (iv). One has successively

$$\begin{aligned} w_\alpha(X)tw_\alpha(X)^{-1} &= \exp(X) \exp(-X^{-1}) \exp(X)t \exp(-X) \exp(X^{-1}) \exp(-X) \\ &= \exp(X) \exp(-X^{-1}) \exp(X - \alpha(t)X) \exp(\alpha(t)^{-1}X^{-1}) \exp(-\alpha(t)X)t. \end{aligned}$$

⁽²⁵⁾**N.D.E.** The first formula of the original has been corrected.

Applying formula (F), one has

$$\begin{aligned} \exp(-X^{-1}) \exp((1 - \alpha(t))X) &= \exp((\alpha(t)^{-1} - 1)X) \alpha^*(\alpha(t)^{-1}) \\ &\quad \cdot \exp(-\alpha(t)^{-1}X^{-1}). \end{aligned}$$

Substituting in the preceding relation, one finds

$$\begin{aligned} \text{int}(w_\alpha(X))t &= \exp(\alpha(t)^{-1}X) \alpha^*(\alpha(t)^{-1}) \exp(-\alpha(t)X)t \\ &= \exp(aX) \alpha^*(\alpha(t)^{-1})t, \end{aligned}$$

where

$$a = \alpha(t)^{-1} - (\alpha \circ \alpha^*)(\alpha(t)^{-1})\alpha(t),$$

but $\alpha \circ \alpha^* = 2$, which gives at once $a = 0$ and $w_\alpha(X) \in N^\times(S')$.

Let us now prove the second assertion of (iv). One has⁽²⁶⁾

$$\begin{aligned} \alpha^*(z)w_\alpha(X) &= \exp(z^2X) \exp(-z^{-2}X^{-1}) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp((z^2 - z)X) \exp(-z^{-2}X^{-1}) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp(-z^{-1}X^{-1}) \alpha^*(z)^{-1} \exp((z^3 - z^2)X) \exp(z^2X) \alpha^*(z) \\ &= \exp(zX) \exp(-z^{-1}X^{-1}) \exp(zX) = w_\alpha(zX). \end{aligned}$$

Let us prove (v). By the preceding result, the first formula of (v) follows at once from the second; let us prove the latter:

$$\begin{aligned} w_\alpha(X)^2 &= \exp(X) \exp(-X^{-1}) \exp(2X) \exp(-X^{-1}) \exp(X) \\ &= \exp(X) \exp(-X^{-1}) \exp(X^{-1}) \alpha^*(-1) \exp(-2X) \exp(X) \\ &= \exp(X) \alpha^*(-1) \exp(-X) = \alpha^*(-1), \end{aligned}$$

since $\alpha(\alpha^*(-1)) = (-1)^2 = 1$, which proves that $\alpha^*(-1) \in \text{Centr}(G)(S)$.

Finally, let us prove (vi). The first assertion is a particular case of the second; let us prove the latter. Both sides of this formula define morphisms from $W(\mathfrak{g}^\alpha)^\times \times_S W(\mathfrak{g}^{-\alpha})^\times$ to G . To prove that they coincide, it suffices to do so on an open subset nonempty on each fiber (Exp. XVIII 1.4); it therefore suffices to check the relation when $1 + XY$ is invertible. One then has successively:

$$\begin{aligned} w_\alpha(X)w_{-\alpha}(Y) &= \exp(X) \exp(-X^{-1}) \exp(X) \exp(Y) \exp(-Y^{-1}) \exp(Y) \\ &= \exp(X) \exp(-X^{-1}) \exp\left(\frac{Y}{1+XY}\right) \alpha^*(1+XY) \\ &\quad \exp\left(\frac{X}{1+XY}\right) \exp(-Y^{-1}) \exp(Y) \\ &= \exp(X) \exp\left(\frac{-X^{-1}}{1+XY}\right) \alpha^*(1+XY) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \exp(-X^{-2}Y^{-1}) \alpha^*\left(\frac{XY}{1+XY}\right) \exp(X+Y^{-1}) \\ &\quad \alpha^*(1+XY) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \exp(-X^{-2}Y^{-1}) \alpha^*(XY) \exp\left(\frac{Y^{-1}+X}{(1+XY)^2}\right) \exp\left(\frac{-Y^{-1}}{1+XY}\right) \exp(Y) \\ &= \alpha^*(XY) \exp(-Y) \exp(Y) = \alpha^*(XY). \end{aligned}$$

⁽²⁶⁾**N.D.E.** The first equality follows from 1.5 (i), which, combined with $\alpha \circ \alpha^* = 2$, gives the formulas

$$\begin{aligned} \alpha^*(z) \exp(X) \alpha^*(z)^{-1} &= \exp(z^2X), \\ \alpha^*(z) \exp(X^{-1}) \alpha^*(z)^{-1} &= \exp(z^{-2}X), \end{aligned}$$

the third equality follows from formula (F), and the fourth from (†), again. Finally, a similar calculation shows that $w_\alpha(X) \alpha^*(z^{-1}) = w_\alpha(zX)$.

Corollary 3.2. *Let $n \in \mathbf{Z}$, $n \neq 0$. For every $w \in G(S)$, the following conditions are equivalent:*

- (i) $w \in N^\times(S)$,
- (ii) *one has $\text{int}(w) \circ n\alpha^* = -n\alpha^*$ (recall that $(n\alpha^*)(z) = \alpha^*(z)^n$).*

One has (i) $\Rightarrow$ (ii) (assertion (iii) of theorem 3.1); conversely, one may suppose that $N^\times$ has a section, and one is reduced to proving:

Lemma 3.3. *One has $\underline{\text{Centr}}_G(n\alpha^*) = T$ for $n \neq 0$.*

Indeed, the image T' of $n\alpha^*$ is a subtorus of G . It follows (Exp. XIX 2.8) that $\underline{\text{Centr}}_G(n\alpha^*)$ is a reductive subgroup of G , containing T . Since on each fiber one has $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) \neq G_{\bar{s}}$, one has $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) = T_{\bar{s}}$ (Exp. XIX 1.6.3⁽²⁷⁾), hence $\underline{\text{Centr}}_G(n\alpha^*) = T$, since these are smooth subgroups of G .

Remark 3.4. The construction of w_α and the fact that $w_\alpha(X)$ normalizes T rely only on formula (F). In particular, if G is an S -group satisfying the conditions of 2.2, $\underline{\text{Norm}}_G(T)$ is different from T on each fiber. It follows that if G is an affine S -group with connected fibers satisfying the conditions of 2.2, it is reductive of semisimple rank 1. Indeed, it is smooth in a neighborhood of the unit section, hence smooth, and one can apply the criterion of Exp. XIX 1.11.

3.5. — Before stating the following theorem, let us make a few remarks. As usual, we identify $\mathfrak{g}^{-\alpha}$ with $(\mathfrak{g}^\alpha)^{\otimes -1}$. Similarly, we shall identify $\underline{\text{Hom}}_{\mathcal{O}_S}(\mathfrak{g}^{-\alpha}, \mathfrak{g}^\alpha)$ with $(\mathfrak{g}^\alpha)^{\otimes 2}$, and hence

$$\underline{\text{Isom}}_{\mathcal{O}_S\text{-mod.}}(\mathbf{W}(\mathfrak{g}^{-\alpha}), \mathbf{W}(\mathfrak{g}^\alpha)) \simeq \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times.$$

If $w \in N^\times(S)$, $\text{Ad}(w)$ interchanges $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$ (3.1, (iii)), hence defines an isomorphism

$$a_\alpha(w) : \mathfrak{g}^{-\alpha} \xrightarrow{\sim} \mathfrak{g}^\alpha,$$

which we shall therefore identify with a section $a_\alpha(w) \in \Gamma(S, (\mathfrak{g}^\alpha)^{\otimes 2})^\times$. This construction is compatible with base change and therefore defines a morphism

$$a_\alpha : N^\times \longrightarrow \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times,$$

such that $a_\alpha(w)Y = \text{Ad}(w)Y$ for all $w \in N^\times(S')$, $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})^\times$, $S' \longrightarrow S$.

Theorem 3.6.

- (i) *One has*

$$\text{int}(w) \exp(Y) = \exp(a_\alpha(w)Y)$$

for every $S' \longrightarrow S$ and all $w \in N^\times(S')$, $Y \in \mathbf{W}(\mathfrak{g}^{-\alpha})(S')$.

- (ii) *One has*

$$a_\alpha(tw) = \alpha(t)a_\alpha(w), \quad a_\alpha(wt) = \alpha(t)^{-1}a_\alpha(w).$$

- (iii) *If one similarly defines $a_{-\alpha} : N^\times \longrightarrow \mathbf{W}((\mathfrak{g}^{-\alpha})^{\otimes 2})^\times$, one has*

$$a_{-\alpha}(w) = a_\alpha(w)^{-1}.$$

(28)

- (iv) *For every $X \in \mathbf{W}(\mathfrak{g}^\alpha)^\times(S')$, $S' \longrightarrow S$, one has*

$$a_\alpha(w_\alpha(X)) = -X^2.$$

⁽²⁷⁾**N.D.E.** The hypothesis $\underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) \neq G_{\bar{s}}$ implies that $\dim \underline{\text{Centr}}_{G_{\bar{s}}}(n\alpha_{\bar{s}}^*) - \dim T_{\bar{s}} < 2$, whereas this difference is even, by loc. cit.

⁽²⁸⁾**N.D.E.** I.e. $a_{-\alpha}(w)$ and $a_{-\alpha}(w)$ are paired, cf. 2.6.1.

Assertion (i) is trivial by the characterization of the morphisms $\exp$ given in 1.5. Assertion (ii) is immediate, as is (iii). Let us prove (iv): let $X \in \Gamma(S', \mathfrak{g}^\alpha)^\times$, $Z \in \Gamma(S', \mathfrak{g}^\alpha)$; by definition, one has⁽²⁹⁾

$$\begin{aligned} a_\alpha(w_\alpha(X))^{-1}(Z) &= \text{Ad}(w_\alpha(X))(Z) \\ &= \text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1})) \text{Ad}(\exp(X))(Z). \end{aligned}$$

Applying formulas (2') and (2) of lemma 2.10, as well as the equalities $H_{-\alpha} = -H_\alpha$, $\bar{\alpha}(H_\alpha) = 2$ (loc. cit. (4) and (5)) and $\langle X, X^{-1} \rangle = 1$ (2.6), one obtains that the right-hand term equals, successively:

$$\begin{aligned} \text{Ad}(\exp(X)) \text{Ad}(\exp(-X^{-1}))(Z) &= \text{Ad}(\exp(X))(Z + \langle X^{-1}, Z \rangle (H_\alpha - X^{-1})) \\ &= Z + \langle X^{-1}, Z \rangle (H_\alpha - 2X - X^{-1} - H_\alpha + X) \\ &= Z - \langle X^{-1}, Z \rangle X - \langle X^{-1}, Z \rangle X^{-1}. \end{aligned}$$

But $Z = \langle X^{-1}, Z \rangle X$ and $\langle X^{-1}, Z \rangle X^{-1} = X^{-2}Z$, hence this shows that $a_\alpha(w_\alpha(X))^{-1} = -X^{-2}$, whence $a_\alpha(w_\alpha(X)) = -X^2$.

Corollary 3.7. *In particular, one has*

$$\text{int}(w_\alpha(X)) \exp(X) = \exp(-X^{-1}),$$

whence (by the definition of $w_\alpha(X)$):

$$w_\alpha(X) \exp(X) w_\alpha(X)^{-1} = \exp(-X) w_\alpha(X) \exp(-X),$$

i.e. by an immediate calculation,

$$(w_\alpha(X) \exp(X))^3 = e.$$

Corollary 3.8. *Let $X \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ and $n \in \mathbf{Z}$, $n \neq 0$. Then $w_\alpha(X)$ is the unique section $w \in G(S)$ satisfying*

- (i) $\text{int}(w) \circ n\alpha^* = -n\alpha^*$.
- (ii) $(w \exp(X))^3 = e$.

One knows that $w_\alpha(X)$ indeed satisfies these conditions. Conversely, let $w \in G(S)$ satisfy (i) and (ii). By 3.2 and 3.1 (ii), one knows that there exists $t \in T(S)$ such that $w = w_\alpha(X)t$. Put $u = \exp(X)$. One then has

$$wuw^{-1} = w_\alpha(X)t \exp(X)t^{-1}w_\alpha(X)^{-1} = \exp(-\alpha(t)X^{-1}),$$

and, on the other hand,

$$\begin{aligned} u^{-1}wu^{-1} &= \exp(-X)w_\alpha(X)t \exp(-X) \\ &= \exp(-X)w_\alpha(X) \exp(-X) \exp(X - \alpha(t)X)t \\ &= \exp(-X^{-1}) \exp(X - \alpha(t)X)t = \exp(-X^{-1})t \exp(H). \end{aligned}$$

Now $(wu)^3 = e \iff wuw^{-1} = u^{-1}wu^{-1}$; comparing the two decompositions of this element on $U_{-\alpha} \cdot T \cdot U_\alpha$, one obtains $t = e$.

Remark 3.9. One can summarize a number of the results of this section by the following diagram of principal homogeneous bundles (on the left):

$$\begin{array}{ccc} \mathbf{W}(\mathfrak{g}^\alpha)^\times & \xrightarrow{w_\alpha} & N^\times \xrightarrow{a_\alpha} \mathbf{W}((\mathfrak{g}^\alpha)^{\otimes 2})^\times \\ & & \\ \mathbf{G}_{\mathbf{m},S} & \xrightarrow{\alpha^*} & T \xrightarrow{\alpha} \mathbf{G}_{\mathbf{m},S}. \end{array}$$

⁽²⁹⁾**N.D.E.** The following has been expanded.

Note that a_α is faithfully flat (since α is) and that w_α is a monomorphism if and only if α^* is a monomorphism. We leave it to the reader to write the corresponding diagrams for the structures of principal bundles on the right, as well as diagrams of the same kind for the root $-\alpha$, and to study the relations between these various diagrams.

Lemma 3.10. *Let S be a scheme, q an integer > 0 such that $x \mapsto x^q$ defines an endomorphism of $\mathbf{G}_{a,S}$, (G, T, α) and (G', T', α') two elementary S -systems, $f : G \rightarrow G'$ a morphism of S -groups. Let*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{\alpha'}$$

be an isomorphism of $\mathcal{O}_S$ -modules, and

$$h^\vee : (\mathfrak{g}^{-\alpha})^{\otimes q} \xrightarrow{\sim} \mathfrak{g}'^{-\alpha'}$$

the contragredient isomorphism. For every $S' \rightarrow S$ and every $X \in W(\mathfrak{g}^\alpha)(S')$, suppose that:

$$f(\exp(X)) = \exp(h(X^q)).$$

Then the following conditions are equivalent:

- (i) $f(\alpha^*(z)) = \alpha'^*(z)^q$.
- (ii) $f(w_\alpha(Z)) = w_{\alpha'}(h(Z^q))$.
- (iii) $f(\exp(Y)) = \exp(h^\vee(Y^q))$.

(Each condition is to be read: for every $S' \rightarrow S$ and every $z \in \mathbf{G}_m(S')$, $Z \in W(\mathfrak{g}^\alpha)^\times(S')$, $Y \in W(\mathfrak{g}^{-\alpha})(S')$, one has ...)

Indeed, (i) $\Rightarrow$ (ii) by 3.8, (ii) $\Rightarrow$ (iii) by 3.7, (iii) $\Rightarrow$ (i) by 2.7.

Proposition 3.11. *Let S be a scheme, $q \in \mathbf{Z}$, $q > 0$, such that $x \mapsto x^q$ defines an endomorphism of $\mathbf{G}_{a,S}$, (G, T, α) and (G', T', α') two elementary S -systems, $f : G \rightarrow G'$ a morphism of S -groups. The following conditions on f are equivalent:*

- (i) *The restriction of f to T factors as a morphism $f_T : T \rightarrow T'$ making the diagram commute:*

$$\begin{array}{ccccc} \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\ q \downarrow & & \downarrow f_T & & \downarrow q \\ \mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S} \end{array}$$

- (ii) *There exists a (unique) isomorphism of $\mathcal{O}_S$ -modules*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \rightarrow \mathfrak{g}'^{\alpha'}$$

such that

$$f(\exp(X)) = \exp(h(X^q)), \quad f(\exp(Y)) = \exp(h^\vee(Y^q))$$

for all $X \in W(\mathfrak{g}^\alpha)(S')$, $Y \in W(\mathfrak{g}^{-\alpha})(S')$, $S' \rightarrow S$ (it follows that f also satisfies the equivalent conditions of 3.10).

One has (ii) $\Rightarrow$ (i). Indeed, by 3.10, condition (ii) implies $f \circ \alpha^* = q\alpha'^*$, hence, by 3.3, $f|_T$ factors through T' . It remains to prove that $\alpha'(f(t)) = \alpha(t)^q$, which follows at once from the fact that f induces a morphism of groups $T \cdot U_\alpha \rightarrow T' \cdot U_{\alpha'}$.

Let us prove (i) $\Rightarrow$ (ii). Let $X \in \Gamma(S, \mathfrak{g}^\alpha)$, $Y \in \Gamma(S, \mathfrak{g}^{-\alpha})$. Put $p_+(x) = f(\exp(xX))$ and $p_-(x) = f(\exp(yY))$; these are morphisms of groups

$$p_+, p_- : \mathbf{G}_{a,S} \rightarrow G.$$

Now one has

$$\begin{aligned} \text{int}(\alpha'^*(z))^q(p_+(x)) &= \text{int}(f_T(\alpha^*(z)))(f(\exp(xX))) \\ &= f(\text{int}(\alpha^*(z))(\exp(xX))) \\ &= f(\exp(z^2xX)) = p_+(z^2x). \end{aligned}$$

Applying lemma 1.2 (with $Q = \alpha'^*(\mathbf{G}_{m,S})$), one deduces that there exists a section $X' \in \Gamma(S, \mathfrak{g}^{\alpha'})$ such that

$$f(\exp(xX)) = p_+(x) = \exp(x^q X').$$

Similarly, there exists a section $Y' \in \Gamma(S, \mathfrak{g}^{-\alpha'})$ such that

$$f(\exp(yY)) = \exp(y^q Y').$$

Writing now that f is a morphism of groups, hence respects formula (F), one obtains at once

$$X^q Y^q = (XY)^q = X' Y'.$$

One easily concludes that $X^q \mapsto X'$ and $Y^q \mapsto Y'$ define isomorphisms h and $h^\vee$ as announced.

Proposition 3.12. *Let (G, T, α) be an elementary S -system, $w \in N^\times(S)$; put*

$$\Omega_0 = \Omega \cap \text{int}(w^{-1})(\Omega).$$

Let d be the function on Ω defined by

$$d(\exp(Y) \cdot t \cdot \exp(X)) = \alpha(t)^{-1} + XY.$$

Then $\Omega_0 = \Omega_d$, and for $\exp(Y) \cdot t \cdot \exp(X) \in \Omega_0(S')$ one has the following formula (put $z = d(\exp(Y) \cdot t \cdot \exp(X))$):

$$(\star) \quad \begin{aligned} \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(z^{-1} a_\alpha(w)^{-1} X) \\ &\quad \cdot t \alpha^*(z) \cdot \exp(z^{-1} a_\alpha(w) Y). \end{aligned}$$

Moreover, $d \circ \text{int}(w) = d^{-1}$.

Indeed, one has at once⁽³⁰⁾

$$\begin{aligned} \text{int}(w)(\exp(Y) \cdot t \cdot \exp(X)) &= \exp(a_\alpha(w) Y) \cdot t \alpha^*(\alpha(t)^{-1}) \cdot \exp(a_\alpha(w)^{-1} X) \\ &= \exp(a_\alpha(w) Y) \cdot \exp(\alpha(t) a_\alpha(w)^{-1} X) \cdot t \alpha^*(\alpha(t)^{-1}). \end{aligned}$$

By 2.1, this is a section of Ω if and only if $1 + \alpha(t)XY$ is invertible, which indeed proves $\Omega_0 = \Omega_d$; then applying formula (F) of loc. cit., one deduces by an immediate calculation the announced formula $(\star)$. Finally, it follows from $(\star)$ that one has

$$\begin{aligned} (d \circ \text{int}(w))(\exp(Y) \cdot t \cdot \exp(X)) &= \alpha(t \alpha^*(z))^{-1} + z^{-2} XY \\ &= z^{-2} (\alpha(t)^{-1} + XY) = z^{-1}, \end{aligned}$$

whence the last assertion.

N. B. One will note that the function d is independent of the choice of w .

4. The isomorphism theorem

Theorem 4.1. *Let S be a scheme, $q \in \mathbf{Z}$, $q > 0$, such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$, (G, T, α) and (G', T', α') two elementary S -systems. Let*

$$h : (\mathfrak{g}^\alpha)^{\otimes q} \longrightarrow \mathfrak{g}'^{\alpha'} \quad \text{and} \quad h^\vee : (\mathfrak{g}^{-\alpha})^{\otimes q} \longrightarrow \mathfrak{g}'^{-\alpha'}$$

⁽³⁰⁾**N.D.E.** The original has been corrected by interchanging $a_\alpha(w)$ and $a_\alpha(w)^{-1}$, and the proof of $d \circ \text{int}(w) = d^{-1}$ has been expanded.

be two mutually contragredient isomorphisms. Let $f_T : T \rightarrow T'$ be a morphism of S -groups making the diagram commute:

$$\begin{array}{ccc} \mathbf{G}_{m,S} & \xrightarrow{q} & \mathbf{G}_{m,S} \\ \alpha^* \downarrow & & \downarrow \alpha'^* \\ T & \xrightarrow{f_T} & T' \\ \alpha \downarrow & & \downarrow \alpha' \\ \mathbf{G}_{m,S} & \xrightarrow{q} & \mathbf{G}_{m,S}. \end{array}$$

There exists a unique morphism of S -groups $f : G \rightarrow G'$ extending f_T and satisfying

$$f(\exp(X)) = \exp(h(X^q))$$

for every $X \in W(\mathfrak{g}^\alpha)(S')$, $S' \rightarrow S$. Moreover, this morphism also satisfies

$$f(\exp(Y)) = \exp(h^\vee(Y^q)) \quad \text{and} \quad f(w_\alpha(Z)) = w_\alpha(h(Z^q)),$$

for every $S' \rightarrow S$ and all $Y \in \Gamma(S', \mathfrak{g}^{-\alpha})$, $Z \in \Gamma(S', \mathfrak{g}^\alpha)^\times$.

If $f : G \rightarrow G'$ extends f_T , then $f \circ \alpha^* = (\alpha'^*)^q$. If in addition f satisfies the second condition, it also satisfies the other two by 3.10. It follows that f is determined on Ω by the relation

$$f(\exp(Y)t\exp(X)) = \exp(h^\vee(Y^q))f_T(t)\exp(h(X^q)).$$

Since Ω is schematically dense in G , this already proves the uniqueness of f . To prove its existence, it suffices, by Exp. XVIII 2.3, to prove that the preceding formula defines a “generically multiplicative” morphism from Ω to G' . Now, by 2.4, this amounts to checking that $\alpha' \circ f = \alpha^q$, which follows from the fact that f extends f_T .

Scholium 4.2. One can also interpret 4.1 as follows: consider the category $\mathcal{E}$ of elementary S -systems and the category $\mathcal{D}$ of pairs

$$\left(\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathcal{L} \right),$$

where T is a torus, α and α^* morphisms of groups such that $\alpha \circ \alpha^* = 2$, and $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module (the reader will specify the morphisms of the two categories considered). One defines a functor $\mathcal{E} \rightarrow \mathcal{D}$ by

$$(G, T, \alpha) \mapsto \left(\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathfrak{g}^\alpha \right).$$

The preceding theorem says that this functor is fully faithful. It is in fact an equivalence of categories, as will be seen in the next section. One already has:

Corollary 4.3. If $q = 1$ and f_T is an isomorphism, then f is an isomorphism.

Corollary 4.4. If $q = 1$ and f_T is faithfully flat with kernel Q (cf. Exp. IX 2.7), then f is faithfully flat (quasi-compact) with kernel Q , hence identifies G' with G/Q .

Indeed, if f_T is faithfully flat with kernel Q , then

$$Q = \text{Ker}(f_T) \subset \text{Ker}(f_T \circ \alpha') = \text{Ker}(\alpha).$$

Introducing the elementary S -system $(G/Q, T/Q, r/Q)$ of 2.13, one is reduced by 2.14 to proving that f/Q induces an isomorphism from G/Q onto G' , which follows at once from 4.3.

5. Examples of elementary systems, applications

5.1. — Let S be a scheme, $\mathcal{L}$ an invertible $\mathcal{O}_S$ -module. Consider the group $G_{\mathcal{L}}$ over S defined by

$$G_{\mathcal{L}}(S') = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left| \begin{array}{l} a, d \in \mathbf{G}_a(S'), \quad b \in W(\mathcal{L})(S'), \\ c \in W(\mathcal{L}^{-1})(S'), \quad ad - bc \in \mathbf{G}_m(S') \end{array} \right. \right\},$$

equipped with the usual matrix multiplication law. It is locally isomorphic to $\mathrm{GL}_{2,S}$. It is therefore a group S -scheme, affine and smooth over S , with connected fibers.

Remark. Let $\mathcal{L}'$ and $\mathcal{L}''$ be two invertible sheaves on S such that $\mathcal{L} = \mathcal{L}' \otimes \mathcal{L}''^{-1}$.⁽³¹⁾ Then one has an isomorphism of S -groups:

$$G_{\mathcal{L}} \xrightarrow{\sim} \mathrm{GL}(\mathcal{L}' \oplus \mathcal{L}'')$$

defined as follows: if x (resp. y) is a section of $\mathcal{L}'$ (resp. $\mathcal{L}''$) over an open subset V of S , one has

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

5.2. — One denotes by $S_{\mathcal{L}}$ the closed subgroup of $G_{\mathcal{L}}$ defined by $ad - bc = 1$. It is also a group S -scheme, affine and smooth over S , with connected fibers (isomorphic to $\mathrm{SL}(\mathcal{L}' \oplus \mathcal{L}'')$ by the preceding isomorphism).

Similarly, consider the morphism $\mathbf{G}_{m,S} \rightarrow G_{\mathcal{L}}$ defined by $z \mapsto \begin{pmatrix} z & 0 \\ 0 & z \end{pmatrix}$. It is a central monomorphism; passing to the quotient, one deduces a group $P_{\mathcal{L}}$, smooth and affine over S , with connected fibers (cf. Exp. VIII 5.7).

One can see that, passing to the quotient from the isomorphism of the preceding remark, $P_{\mathcal{L}}$ is identified with the group of automorphisms of the projective bundle $\mathbf{P}(\mathcal{L}' \oplus \mathcal{L}'')$ (cf. EGA, II 4.2.7). One denotes by i and p the canonical morphisms

$$S_{\mathcal{L}} \xrightarrow{i} G_{\mathcal{L}} \xrightarrow{p} P_{\mathcal{L}};$$

i is a closed immersion, p is faithfully flat and affine.

5.3. — Consider the morphisms of groups

$$\begin{aligned} t_G : \mathbf{G}_{m,S}^2 &\rightarrow G_{\mathcal{L}}, & t_G(z, z') &= \begin{pmatrix} z & 0 \\ 0 & z' \end{pmatrix}; \\ t_S : \mathbf{G}_{m,S} &\rightarrow S_{\mathcal{L}}, & t_S(z) &= \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}; \\ t_P : \mathbf{G}_{m,S} &\rightarrow P_{\mathcal{L}}, & t_P(z) &= p(t_G(z, 1)). \end{aligned}$$

These are monomorphisms of groups defining in each group a (split) torus of relative codimension 2. For every $s \in S$, let

$$X \in \Gamma(\bar{s}, \mathcal{L} \otimes \bar{s})^{\times};$$

then the section $\begin{pmatrix} 0 & X \\ X^{-1} & 0 \end{pmatrix}$ of $G_{\mathcal{L},\bar{s}}$ normalizes $t_G(\mathbf{G}_{m,\bar{s}}^2)$ and does not centralize it; one then concludes from Exp. XIX 1.6 that $G_{\mathcal{L}}$ is *reductive*, of *semisimple rank* 1, with *maximal torus* $t_G(\mathbf{G}_{m,S}^2)$.

One reasons similarly for $S_{\mathcal{L}}$ and $P_{\mathcal{L}}$, and sees that $S_{\mathcal{L}}$ (resp. $P_{\mathcal{L}}$) is *reductive*, of *semisimple rank* 1, with maximal torus $t_S(\mathbf{G}_{m,S})$ (resp. $t_P(\mathbf{G}_{m,S})$).

5.4. — Reasoning as usual, one immediately determines the Lie algebra of these various groups and the adjoint action of the chosen maximal torus. Let us do so for $G_{\mathcal{L}}$; this is immediate by Exp. II 4.8: $\mathrm{Lie}(G_{\mathcal{L}}/S)$ is the Lie algebra of the matrices below:

$$\mathrm{Lie}(G_{\mathcal{L}}/S) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \left| \begin{array}{l} a \text{ and } d \text{ sections of } \mathcal{O}_S, \quad b \text{ a section of } \mathcal{L}, \\ c \text{ a section of } \mathcal{L}^{-1} \end{array} \right. \right\}$$

⁽³¹⁾**N.D.E.** $\mathcal{L}'' \otimes \mathcal{L}'^{-1}$ has been corrected to $\mathcal{L}' \otimes \mathcal{L}''^{-1}$, and the following sentence has been expanded.

with the usual bracket; one has

$$\mathrm{Ad}(t_G(z, z')) \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & zz'^{-1}b \\ z'z^{-1}c & d \end{pmatrix}.$$

Put $\mathcal{L}ie(G_{\mathcal{L}}/S) = \mathfrak{g}$. Let $\alpha_G : t_G(\mathbf{G}_{m,S}^2) \rightarrow \mathbf{G}_{m,S}$ be the character defined by

$$\alpha_G(t_G(z, z')) = zz'^{-1}.$$

One sees at once from the preceding relation that α_G is a root of $G_{\mathcal{L}}$ with respect to $t_G(\mathbf{G}_{m,S}^2)$, and that the morphism

$$u : \mathcal{L} \rightarrow \mathfrak{g} \quad (\text{resp. } u_- : \mathcal{L}^{-1} \rightarrow \mathfrak{g})$$

defined by $u(X) = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}$ (resp. $u_-(X) = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}$) is an isomorphism from $\mathcal{L}$ onto $\mathfrak{g}^{\alpha_G}$ (resp. from $\mathcal{L}^{-1}$ onto $\mathfrak{g}^{-\alpha_G}$).

One has therefore proved that $(G, t_G(\mathbf{G}_{m,S}^2), \alpha_G)$ is an elementary S -system.

Putting similarly

$$\alpha_S(t_S(z)) = z^2, \quad \alpha_P(t_P(z)) = z,$$

one proves that $(S_{\mathcal{L}}, t_S(\mathbf{G}_{m,S}), \alpha_S)$ and $(P_{\mathcal{L}}, t_P(\mathbf{G}_{m,S}), \alpha_P)$ are elementary systems, and defines isomorphisms of $\mathcal{L}$ (resp. $\mathcal{L}^{-1}$) with the corresponding direct summands of the Lie algebras of $S_{\mathcal{L}}$ and $P_{\mathcal{L}}$.

5.5. — Put $\exp \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}$. One has thus defined a morphism

$$W(\mathfrak{g}^{\alpha_G}) \rightarrow G_{\mathcal{L}}$$

which induces on Lie algebras the canonical morphism, hence is the unique morphism of this type (1.5). Similarly, put $\exp \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}$. Calculating formula (F) explicitly, one finds

$$\left\langle \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ Y & 0 \end{pmatrix} \right\rangle = XY, \quad \alpha_G^*(z) = \begin{pmatrix} z & 0 \\ 0 & 1/z \end{pmatrix} = t_G(z, z^{-1}).$$

⁽³²⁾The open subset $N^\times = N_G^\times$ (defined before 3.1) is:

$$N_G^\times(S') = \left\{ \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \mid P \in W(\mathfrak{g}^\alpha)^\times(S'), Q \in W(\mathfrak{g}^{-\alpha})^\times(S') \right\},$$

the morphism w_{α_G} (cf. 3.1 (iv)) is given, for every $X \in W(\mathfrak{g}^\alpha)^\times(S')$, by

$$w_{\alpha_G}(X) = \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix};$$

the morphism a_{α_G} (cf. 3.5) is given by:

$$\text{if } w = \begin{pmatrix} 0 & P \\ Q & 0 \end{pmatrix} \in N_G^\times(S'), \quad \text{then } a_{\alpha_G}(w) = PQ^{-1} \in W((\mathfrak{g}^\alpha)^{\otimes 2})^\times(S'),$$

i.e. for every $Y \in W(\mathfrak{g}^{-\alpha})^\times(S')$, one has $a_{\alpha_G}(w)(Y) = PQ^{-1}Y \in W(\mathfrak{g}^\alpha)^\times(S')$.

5.6. — We leave it to the reader to perform the same calculations in $S_{\mathcal{L}}$ and $P_{\mathcal{L}}$. One finds the same duality formula and the coroots

$$\alpha_S^*(z) = t_S(z), \quad \alpha_P^*(z) = t_P(z^2).$$

Let p_T be the morphism induced by $p : G_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$ on $t_S(\mathbf{G}_{m,S})$, i.e.

$$p_T(t_S(z)) = t_P(z^2).$$

⁽³²⁾**N.D.E.** The following has been expanded.

One therefore has the commutative diagram:⁽³³⁾

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \text{id} & \searrow \alpha_S^* & \swarrow \alpha_P^* & \searrow 2 \\
 \mathbf{G}_{m,S} & \xrightarrow{t_S} & t_S(\mathbf{G}_{m,S}) & \xrightarrow{p_T} & t_P(\mathbf{G}_{m,S}) \xrightarrow{t_P} \mathbf{G}_{m,S} \\
 & \searrow 2 & \swarrow \alpha_S & \swarrow \alpha_P & \searrow \text{id} \\
 & & \mathbf{G}_{m,S} & &
 \end{array}$$

One recognizes in the central part the commutative diagram of 4.1⁽³⁴⁾ relative to the canonical morphism $p \circ i : S_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$, which induces a morphism of the preceding elementary S -systems.

5.7. — Let now (G, T, α) be any elementary S -system. Consider the commutative diagram:

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \text{id} & \downarrow \alpha^* & \searrow 2 & \\
 \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\
 & \searrow 2 & \downarrow \alpha & \swarrow \text{id} & \\
 & & \mathbf{G}_{m,S} & &
 \end{array}$$

Combining the preceding two diagrams, one obtains a commutative diagram:

$$\begin{array}{ccccc}
 & & \mathbf{G}_{m,S} & & \\
 & \swarrow \alpha_S^* & \downarrow \alpha^* & \swarrow \alpha_P^* & \\
 t_S(\mathbf{G}_{m,S}) & \xrightarrow{\alpha^* \circ t_S^{-1}} & T & \xrightarrow{t_P \circ \alpha} & t_P(\mathbf{G}_{m,S}) \\
 & \searrow \alpha_S & \downarrow \alpha & \swarrow \alpha_P & \\
 & & \mathbf{G}_{m,S} & &
 \end{array}$$

Using 4.1, one therefore has:

Proposition 5.8. *Let S be a scheme, (G, T, α) an elementary S -system. Put $\mathcal{L} = \mathfrak{g}^\alpha$ (and hence $\mathcal{L}^{-1} = \mathfrak{g}^{-\alpha}$).*

(i) *There exists a unique morphism of groups $f : S_{\mathcal{L}} \rightarrow G$ satisfying the following equivalent conditions:*

$$\begin{aligned}
 (a) \quad & f \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} = \alpha^*(z), & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X); \\
 (b) \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix} = \exp(Y); \\
 (c) \quad & f \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix} = \exp(X), & f \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix} = w_\alpha(X).
 \end{aligned}$$

(ii) *There exists a unique morphism of groups $g : G \rightarrow P_{\mathcal{L}}$ satisfying*

$$g(t) = \begin{pmatrix} \alpha(t) & 0 \\ 0 & 1 \end{pmatrix}, \quad g(\exp(X)) = p \begin{pmatrix} 1 & X \\ 0 & 1 \end{pmatrix}.$$

⁽³³⁾**N.D.E.** Where t_S and t_P are isomorphisms.

⁽³⁴⁾**N.D.E.** With $q = 1$.

Moreover, one has

$$g(\exp(Y)) = p \begin{pmatrix} 1 & 0 \\ Y & 1 \end{pmatrix}, \quad g(w_\alpha(X)) = p \begin{pmatrix} 0 & X \\ -X^{-1} & 0 \end{pmatrix}.$$

The morphism g is faithfully flat quasi-compact with kernel $\text{Ker}(\alpha) = \underline{\text{Centr}}(G)$, and $g \circ f$ is the canonical morphism $S_{\mathcal{L}} \rightarrow P_{\mathcal{L}}$.

Note that conditions (b) of (i) give an explicit description of the duality between $\mathfrak{g}^\alpha$ and $\mathfrak{g}^{-\alpha}$.

Corollary 5.9. *Let (G, T, α) be an elementary S -system. The subgroups $T \cdot U_\alpha$, $T \cdot U_{-\alpha}$, U_α and $U_{-\alpha}$ are closed.*

Since U_α is a closed subgroup scheme of $T \cdot U_\alpha$, it suffices to check this for the latter. By Noether's theorem (Exp. IV 5.3.1 and 6.4.1), it suffices to prove that $(T \cdot U_\alpha)/\text{Ker}(\alpha)$ is a closed subgroup of $G/\text{Ker}(\alpha)$. By 5.8, one is therefore reduced to proving that the subgroup of $P_{\mathcal{L}}$ (or of $G_{\mathcal{L}}$, which amounts to the same thing by another application of Noether's theorem) defined by $c = 0$ is closed, which is trivial.

Consequently, the morphisms $\exp$ of theorem 1.5 (i) are closed immersions.

N. B. The corollary also follows from the fact that $T \cdot U_\alpha$ and $T \cdot U_{-\alpha}$ are “Borel subgroups” of G (cf. Exp. XII 7.10).

5.10. — Let $\mathcal{L}$ be an invertible $\mathcal{O}_S$ -module and

$$\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}$$

a diagram of groups⁽³⁵⁾ such that $\alpha \circ \alpha^* = 2$. Let R be the maximal torus of $\text{Ker}(\alpha)$ and $K = \alpha^{*-1}(R)$. Then K is a subgroup of multiplicative type of $\mathbf{G}_{m,S}$; by $\alpha \circ \alpha^* = 2$, it is even a subgroup of $\mu_{2,S}$. In particular, the morphism

$$K \rightarrow S_{\mathcal{L}}, \quad z \mapsto \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}$$

is central. One therefore has a central monomorphism of groups:

$$K \rightarrow R \times S_{\mathcal{L}}, \quad z \mapsto \left(\alpha^*(z), \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} \right).$$

Consider the group $G = (R \times S_{\mathcal{L}})/K$ obtained by passing to the quotient. It is a group affine and smooth over S , with connected fibers. It is immediate that the sequence

$$1 \rightarrow K \rightarrow R \times t_S(\mathbf{G}_{m,S}) \rightarrow T \rightarrow 1$$

where $u(x, t_S(z)) = x\alpha^*(z)$ is exact. The image of $R \times t_S(\mathbf{G}_{m,S})$ in G is therefore a torus T' isomorphic to T . One now shows without difficulty that if α' is the character of T' deduced from α by the preceding isomorphism, (G, T', α') is an elementary S -system, that $\mathfrak{g}^{\alpha'}$ is isomorphic to $\mathcal{L}$, and that α'^* is obtained from α^* by the isomorphism $T \xrightarrow{\sim} T'$. One has therefore constructed an elementary S -system (G, T', α') such that the corresponding object $(\mathbf{G}_{m,S} \xrightarrow{\alpha'^*} T' \xrightarrow{\alpha'} \mathbf{G}_{m,S}, \mathfrak{g}^{\alpha'})$ of the category $\mathcal{D}$ defined in 4.2 is isomorphic to $(\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathcal{L})$. One has therefore proved:

Theorem 5.11. *With the notations of 4.2, the functor*

$$(G, T, \alpha) \mapsto (\mathbf{G}_{m,S} \xrightarrow{\alpha^*} T \xrightarrow{\alpha} \mathbf{G}_{m,S}, \mathfrak{g}^\alpha)$$

is an equivalence of categories between $\mathcal{E}$ and $\mathcal{D}$.

6. Generators and relations for an elementary system

⁽³⁵⁾N.D.E. T being a torus.

6.1. — Let S be a scheme, (G, T, α) an elementary S -system. Let $X \in W(\mathfrak{g}^\alpha)^\times(S)$ and $u = \exp(X)$; we saw in 3.8 that the element $w = w_\alpha(X)$ satisfies, in particular, the relation

$$(wu)^3 = e.$$

⁽³⁶⁾Let s_α denote the automorphism of T induced by $\text{int}(w)$; by theorem 3.1 (iii), for every $S' \rightarrow S$ and $t \in T(S')$, one has

$$s_\alpha(t) = \text{int}(w)(t) = t \cdot \alpha^*(\alpha(t)^{-1}).$$

Theorem 6.2. *Let H be a sheaf of S -groups for (fppf). Let*

$$f_T : T \rightarrow H, \quad f_\alpha : U_\alpha \rightarrow H$$

be morphisms of groups and $h \in H(S)$ a section of H . For there to exist a morphism of groups (necessarily unique)

$$f : G \rightarrow H$$

extending f_T and f_α and satisfying $f(w) = h$, it is necessary and sufficient that the following conditions hold:

(i) *For every $S' \rightarrow S$, every $t \in T(S')$ and every $x \in U_\alpha(S')$, one has*

$$(1) \quad f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(txt^{-1}) = f_\alpha(x^{\alpha(t)}).$$

(in other words, f_T and f_α extend to a morphism of groups from the semidirect product $T \cdot U_\alpha$ to H).

(ii) *For every $S' \rightarrow S$ and every $t \in T(S')$, one has*

$$(2) \quad hf_T(t)h^{-1} = f_T(s_\alpha(t)) = f_T(t \cdot \alpha^*(\alpha(t)^{-1})).$$

(iii) *One has the two relations in $H(S)$:*

$$(3) \quad h^2 = f_T(\alpha^*(-1)),$$

$$(4) \quad (hf_\alpha(u))^3 = e.$$

Proof. Let us write U_α and $U_{-\alpha}$ additively, and their vector structure multiplicatively. If f satisfies the conditions of the statement, one necessarily has, for every $y \in U_{-\alpha}(S')$,

$$f(y) = f(w^{-1}wyw^{-1}w) = hf_\alpha(w^{-1}yw)h^{-1}.$$

Let therefore $f_{-\alpha} : U_{-\alpha} \rightarrow H$ be the morphism defined by

$$(*_1) \quad f_{-\alpha}(y) = hf_\alpha(w^{-1}yw)h^{-1}.$$

It is a morphism of groups. On the other hand, f is determined on the big cell Ω by

$$f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x).$$

This shows the uniqueness of f ; since the conditions of the statement are manifestly necessary, let us show that they are sufficient.

By (4), one has

$$hf_\alpha(u)h^{-1}h^2 = f_\alpha(-u)h^{-1}f_\alpha(-u).$$

Now, by (3) and (1), $h^2 = h^{-2}$ commutes with $f_\alpha(-u)$, which gives

$$hf_\alpha(u)h^{-1} = f_\alpha(-u)hf_\alpha(-u).$$

But, by definition, $hf_\alpha(u)h^{-1} = f_{-\alpha}(wuw^{-1})$; by 3.7, since $u = \exp(X)$ and $w = w_\alpha(X)$, one has

$$(*_2) \quad wuw^{-1} = -\tilde{u},$$

where $\tilde{u}$ denotes the element paired with u . One therefore obtains:

$$(*_3) \quad f_{-\alpha}(-\tilde{u}) = f_\alpha(-u)hf_\alpha(-u).$$

⁽³⁶⁾**N.D.E.** The following sentence has been added.

Let now t be a section of T over a variable $S' \rightarrow S$. Let $\text{int}(f_T(t))$ act on the preceding formula. On the left-hand side, one obtains⁽³⁷⁾

$$\begin{aligned} f_T(t)f_{-\alpha}(-\tilde{u})f_T(t)^{-1} &= f_T(t)hf_{\alpha}(u)h^{-1}f_T(t)^{-1} \\ &= h(h^{-1}f_T(t)h)f_{\alpha}(u)(h^{-1}f_T(t)^{-1}h)h^{-1} \\ &= hf_T(s_{\alpha}(t))f_{\alpha}(u)f_T(s_{\alpha}(t))^{-1}h^{-1} \\ &= hf_{\alpha}(\alpha(s_{\alpha}(t))u)h^{-1} \end{aligned}$$

by (2) and (1); then, since $s_{\alpha}(t) = t \cdot \alpha^*(\alpha(t)^{-1})$ and $\alpha \circ \alpha^* = 2$, this equals

$$hf_{\alpha}(\alpha(t)^{-1}u)h^{-1}.$$

Finally, by $(*_1)$ and $(*_2)$, one has

$$hf_{\alpha}(\alpha(t)^{-1}u)h^{-1} = f_{-\alpha}(\alpha(t)^{-1}uwuw^{-1}) = f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}).$$

The right-hand side of $(*_3)$ gives

$$f_{\alpha}(-\alpha(t)u) \cdot f_T(t)hf_T(t)^{-1}h^{-1} \cdot h \cdot f_{\alpha}(-\alpha(t)u),$$

and since $hf_T(t)^{-1}h^{-1} = f_T(s_{\alpha}(t^{-1})) = f_T(t \cdot \alpha^*(\alpha(t)))$, this equals

$$f_{\alpha}(-\alpha(t)u) \cdot f_T(\alpha^*(\alpha(t))) \cdot h \cdot f_{\alpha}(-\alpha(t)u).$$

Comparing the two expressions obtained, one obtains

$$f_{-\alpha}(-\alpha(t)^{-1}\tilde{u}) = f_{\alpha}(-\alpha(t)u) \cdot f_T(\alpha^*(\alpha(t))) \cdot h \cdot f_{\alpha}(-\alpha(t)u).$$

Since $\alpha : T \rightarrow \mathbf{G}_{m,S}$ is faithfully flat and H is a separated presheaf, one may replace $-\alpha(t)^{-1}$ by an arbitrary section of $\mathbf{G}_{m,S}$ and obtains:

Lemma 6.2.1. *For every $z \in \mathbf{G}_m(S')$, $S' \rightarrow S$, one has*

$$f_{-\alpha}(z\tilde{u}) = f_{\alpha}(z^{-1}u) \cdot f_T(\alpha^*(-z^{-1})) \cdot h \cdot f_{\alpha}(z^{-1}u).$$

Let now $x, y \in \mathbf{G}_a(S')$, $S' \rightarrow S$; suppose y and $(1 + xy)$ invertible. First applying the lemma with $z = y$, one obtains⁽³⁸⁾

$$f_{\alpha}(xu)f_{-\alpha}(y\tilde{u}) = f_{\alpha}((x + y^{-1})u) \cdot f_T(\alpha(-y^{-1})) \cdot h \cdot f_{\alpha}(y^{-1}u).$$

Now $x + y^{-1} = y^{-1}(1 + xy)$. Applying the lemma with $z = y/(1 + xy)$, one finds

$$\begin{aligned} f_{\alpha}((x + y^{-1})u) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) \\ &\quad \cdot h^{-1} \cdot f_T\left(\alpha^*\left(\frac{-y}{1 + xy}\right)\right). \end{aligned}$$

Substituting this in the preceding equality, one obtains

$$\begin{aligned} f_{\alpha}(xu)f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) \\ &\quad \cdot h^{-1} \cdot f_T(\alpha^*(1 + xy)^{-1}) \cdot h \cdot f_{\alpha}(y^{-1}u). \end{aligned}$$

Since $h^{-1}f_T(t)h = f_T(s_{\alpha}(t))$ by (2) (note that $s_{\alpha}^2 = \text{id}$), and since $s_{\alpha} \circ \alpha^* = -\alpha^*$ (cf. 6.2.1), this equals

$$f_{-\alpha}\left(\frac{y}{1 + xy}\tilde{u}\right) f_{\alpha}(-(x + y^{-1})u) \cdot f_T(\alpha^*(1 + xy)) \cdot f_{\alpha}(y^{-1}u).$$

Finally, since for all $x' \in U_{\alpha}(S')$ and $z \in \mathbf{G}_m(S')$, one has

$$f_{\alpha}(x')f_T(\alpha^*(z)) = f_T(\alpha^*(z))f_{\alpha}(z^{-2}x'),$$

⁽³⁷⁾**N.D.E.** The following has been corrected.

⁽³⁸⁾**N.D.E.** The following calculations have been expanded.

one obtains

$$\begin{aligned} f_\alpha(xu)f_{-\alpha}(y\tilde{u}) &= f_{-\alpha}\left(\frac{y}{1+xy}\tilde{u}\right) \cdot f_T(\alpha^*(1+xy)) \cdot f_\alpha\left(\left(\frac{-y^{-1}(1+xy)}{(1+xy)^2} + y^{-1}\right)u\right) \\ &= f_{-\alpha}\left(\frac{y}{1+xy}\tilde{u}\right) \cdot f_T(\alpha^*(1+xy)) \cdot f_\alpha\left(\frac{x}{1+xy}u\right). \end{aligned}$$

One has therefore proved:

Lemma 6.2.2. *Let $S' \rightarrow S$. If $a \in U_\alpha(S')$, $b \in U_{-\alpha}^\times(S')$, and $1+ab \in \mathbf{G}_m(S')$, one has*

$$f_\alpha(a)f_{-\alpha}(b) = f_{-\alpha}\left(\frac{b}{1+ab}\right) f_T(\alpha^*(1+ab)) f_\alpha\left(\frac{a}{1+ab}\right).$$

By schematic density, this formula remains valid when $b \in U_{-\alpha}(S')$, $1+ab$ still being invertible. Consider then the morphism

$$f : \Omega \rightarrow H$$

defined by $f(ytx) = f_{-\alpha}(y)f_T(t)f_\alpha(x)$.

It follows at once from 6.2.2, condition 6.2 (i), and formula (F') of 2.4, that if $g, g' \in \Omega(S')$ and $gg' \in \Omega(S')$, one has $f(gg') = f(g)f(g')$. By Exp. XVIII 2.3 (iii) and 2.4,⁽³⁹⁾ there therefore exists a morphism of groups $G \rightarrow H$ extending f . Denote it also by f ; it answers the question; indeed, it suffices to prove that $f(w_\alpha) = h$. Now $w_\alpha = u \cdot (-\tilde{u}) \cdot u$, whence, by $(*)_3$:⁽⁴⁰⁾

$$f(w_\alpha) = f_\alpha(u)f_{-\alpha}(-\tilde{u})f_\alpha(u) = h.$$

Remark 6.3. We shall supplement these results in Exp. XXIII 3.5.

⁽³⁹⁾**N.D.E.** Note that each geometric fiber of G is connected, for example by 1.1.

⁽⁴⁰⁾**N.D.E.** The original has been simplified by invoking $(*)_3$.